HANDBOOK OF POWDER TECHNOLOGY

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The Handbook presents, in convenient form, existing knowledge in all specialized areas of Powder Technology.

Information that can be used for the design of industrial processes involving the production, handling and processing of particulate materials so far did not exist in a form in which it is readily accessible to design engineers. Scientists responsible for characterizing particulate materials, specifying the requirements of industrial processes, operating plants, or setting up quality-control tests all have similar problems in their fact-finding missions through the scattered and scanty literature. The aim of this handbook is to remedy this deficiency by providing a series of thematic volumes on various aspects of powder technology. Each volume is written as a monograph and can be used independently of other volumes.

Emphasis is placed on setting out the basic concepts of the subject and discussing their applications to the design, selection and operation of equipment of an industrial scale. To ensure timely publication, each volume will be published as soon as the material has been delivered by the authors.

Vol. 1. Particle Size Enlargement (C.E. Capes)
Vol. 2. Fundamentals of Gas-Particle Flow (G. Rudinger)
Vol. 3. Solid-Gas Separation (L. Svarovsky)
Vol. 4. Dust Explosions (P. Field)
Vol. 5. Solid-Liquid Separation Processes and Technology (L. Svarovsky)
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Vol. 7. Dispersing Powders in Liquids (R.D. Nelson)
Vol. 8. Gas Fluidization (M. Pell)

Further volumes are in preparation.
(For further information on Volumes 1-7, see p. 123 of this volume.)

GAS FLUIDIZATION

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Dedication

TO THE GIANTS WHO TAUGHT ME WELL :

Arthur Squires

George Curran

Fred Zenz

Acknowledgements

To make this work possible required the assistance of many people.

My wife Ellen who waited patiently as I spent endless hours toiling at the computer.

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Preface

This is intended to serve as a handbook for the practitioner of fluidization on an industrial scale. There are few plants that do not have a fluid bed process operating somewhere on the site. Yet most engineers have never been formally trained in the subject. College courses often emphasize academic issues rather than industrial needs and if one tries to study the literature, there is an overwelming abundance of correlations and experimental data. Unhappily, too much work is done on small scale equipment under ambient conditions. This leads to good small scale science but not to design procedures that apply to industrial systems.

I have minimized theoretical development of fundamental equations in favor of giving correlations that have been developed for large scale equipment. In some cases, where several correlations might apply, I have chosen one or two that work best based upon my own experience.

The book gives design procedures that should assist the designer and the operator of fluid beds to improve his process and to avoid some of the pitfalls that have befallen others in the early days of fluidization science.

This material serves well as a secondary text to a course on fluidization. After the student is out in the field, reference to its contents will be easy and straightforward. The emphasis on commercial design is a stimulating counterpoint to a strongly academic viewpoint.

Wilmington DE Mel Pell

Chapter 1

INTRODUCTION TO GAS FLUIDIZATION

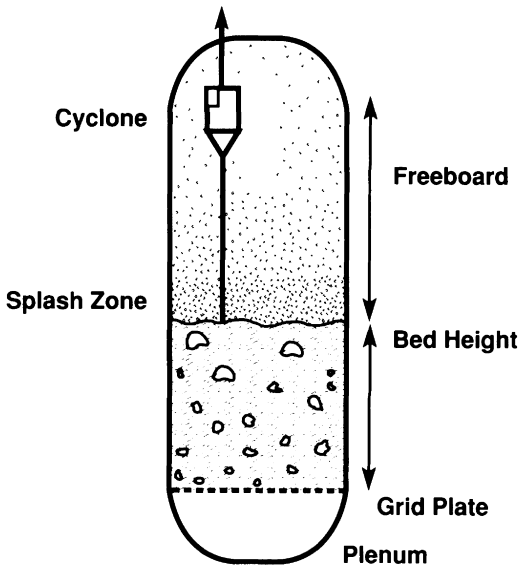
Introduction 1.1

This volume is written from the applications point of view. Theoretical development is minimized and the emphasis is on how to apply principles to working problems. The text discusses gas fluidization but most of the principles apply to liquid fluidized systems as well.

Fluidization is the levitation of a mass (bed) of particles by a fluid. In this state, the particle mass behaves like a fluid in that it tends to establish a level and flow in response to pressure gradients. There is intense mixing and gas solids contact. These properties make it easy to have an isothermal system with good mass transfer. The fluid bed is therefore ideal for reaction, drying, mixing and heat transfer applications.

A few definitions are in order at this point. The device used for the introduction of fluid in the system is called the grid; in some vessels this is a drilled plate on the bottom of the vessel and is called the grid plate. A plenum chamber is usually provided below the grid plate. The level of solids is the bed height and the vertical space above the bed height is called the freeboard (Figure 1–1). A splash zone usually exists as a transition between the bed and freeboard. Cyclones return entrained material to the bed.

Gas flowing through the bed forms voids which look like bubbles in many systems. At sufficiently high velocity, the voids become elongated and lose their semicircular shape but we continue to refer to a bubble phase for convenience. The solids rich part of the bed is called the emulsion phase.



Fluid Bed Definitions Fig. 1–1

Regimes of Fluidization 1.2

As applications of fluid beds have proliferated, it has become evident that there are a number of regimes of fluidization. The fluid bed behaves differently as velocity, gas properties and solids properties are varied.

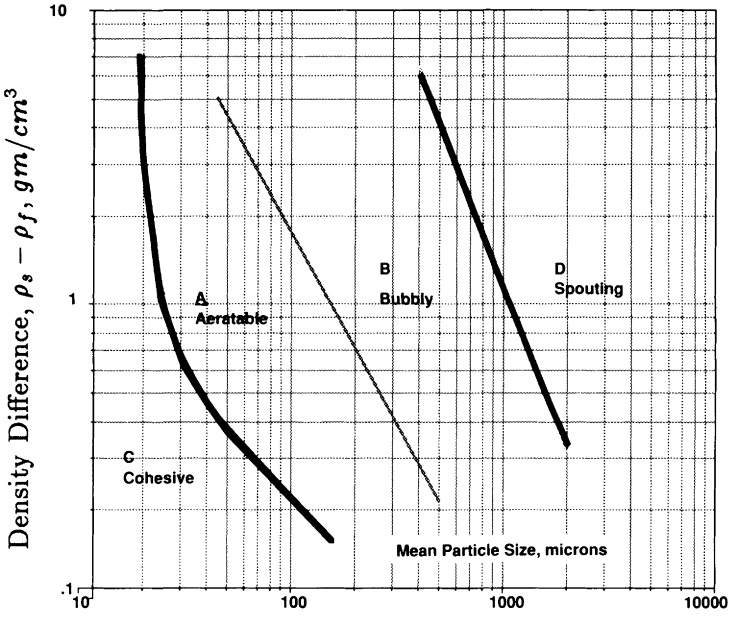
If a bed of solids is exposed to an increasing upflow of gas, the pressure drop increases with gas flow. A point is reached when the pressure drop is enough to balance the weight of the bed. At this point, the bed is fluidized. The velocity at which this occurs is the minimum fluidization velocity, u_{mf}. As the velocity is increased, bubbles appear in the bed. The point at which this occurs is the minimum bubbling velocity, u_{mb}. As the velocity is increased further, the bubbles occupy more and more of the bed. Additionally, the particles tend to be thrown from the bed by the bursting bubbles making the top boundary of the bed indistinct. At some point, the velocity exceeds the terminal velocity of some of the bed particles. The particles can then be entrained from the container even if a long height is provided for fallback of particles from the splash zone.

In order to prevent the bed from becoming depleted as the velocity continues to increase, entrained particles are recovered and returned to the bed. This allows operation at velocities above the terminal velocity of virtually all the bed particles. Beds which have a recognizable surface under these conditions are called turbulent beds. Those which do not have distinct surface are acknowledged to be fast fluid beds. At high velocities such as these, bubbles tend to lose their distinct shape and the gas can become the continuous phase with streamers of solids in gas. The bed level disappears and there is a continuous density gradient from the bottom of the vessel to the top. Eventually the fluid bed becomes an entrained bed in which there is no recognizable bed surface at all and the volume fraction of solids in gas is below 0.1. The exact conditions at which the fluid bed becomes a fast or entrained bed are still subject to debate.

Effects of Particle Size and Density 1.3

In 1973, Geldart (1) published a categorization of particles with respect to how they fluidize. His chart is reproduced in Figure 1–2. Type B particles are in the group called bubbly. They behave like beach sand. Dense materials like glass, sand, ore and light particles over about 150 micrometers (microns) are likely to be type B. The particles do not display any cohesive properties and therefore once the minimum fluidization velocity is exceeded, the excess gas appears in the form of bubbles. For these particles, $u_{mf} = u_{mb}$. Bubbles in a bed of type B particles can grow to considerable size in tall, large diameter systems reaching sizes on the order of a meter.

Type A particles are smaller and lighter on the axes of Figure 1–2 and they are designated as aeratable. Manufactured catalysts generally are in this category with particle sizes of about 20 to 100 microns. These particles form a slightly cohesive structure. As a result, when the velocity is increased beyond u_{mf}, the bed continues to expand smoothly without the formation of bubbles. A point is eventually reached when bubbles start to form. Obviously u_{mb} exceeds u_{mf} and this is one of the characteristics of this material. The small average particle size and the presence of very fine particles promotes bubble splitting and a maximum stable bubble size can be shown to exist (2,3).



Geldart's Particle Classification Scheme Fig. 1–2
redrawn from (1)

On the Geldart diagram, type C particles are even smaller and lighter than type A particles and they are characterized as cohesive. They are usually less than 20 microns. The large external surface area combined with low mass tends to give relatively large attractive affects. This occurs to such an extent that the particles do not flow such as to form bubbles and gas has difficulty in uniformly expanding the bed. Gas flows through the bed in channels called "ratholes". Sometimes the particles will agglomerate into large "static balls" held together by surface forces. A lab scale unit can have the entire bed lift up as a slug of solids. When fluid bed measurements are performed on such a system, a low pressure drop is observed since the gas is flowing in a channel without encountering the particles. Often, type C materials can be fluidized by using high velocity to overcome cohesive effects.

Type D particles are large, on the order of 1 or more millimeters (1000 microns). They differ from group B particles in that as velocity increases, a jet can be formed in the bed and material is then blown out with the jet in a spouting motion. There is considerable bed motion since particles move down and into the jet to replace the entrained material.

Effects of Gas Density and Viscosity 1.4

Note that in Figure 1.2, the ordinate is given as $(\rho_s - \rho_f)$. As the gas density is increased, we tend to move into the type A zone from the type B zone. In general, dense fluid systems such as gases at very high pressure and liquids fluidize like type A systems. As pressure is increased, borderline type B systems behave more like type A systems.

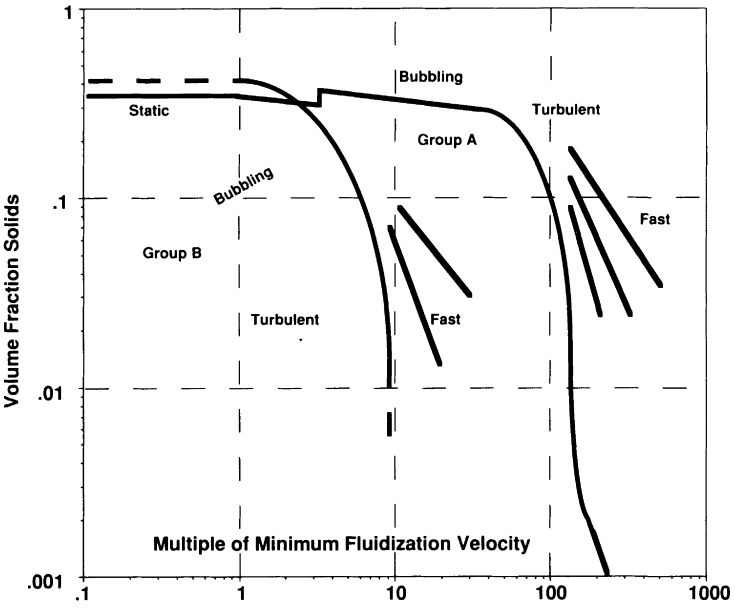
Increasing density or viscosity usually decreases the terminal velocity and the minimum fluidization velocity (4,5). As fluid density and viscosity increase, we move toward aeratable systems with small bubbles and smooth fluidization (6,7). Still, Rowe (8) points out that particles smaller than about 100 microns, where Re_{mf} is less than 2, will have u_{mf} relatively independent of pressure.

Increasing temperature reduces density while increasing viscosity. Usually, u_{mf} will decrease with increasing temperature. But, Botterill and Toeman (9) found that sand particles 2.2 mm in diameter displayed a maximum in u_{mf} with temperature they passed through the transition to turbulent flow.

Squires et al (10) present the effects of velocity on the fluidization regime. In Figure 1–3, the dotted line represents a Geldart type B solid; the solid line is for a type A solid. As velocity increases, the type B solid begins to bubble at the fluidization velocity. The bubbling bed becomes turbulent at about 10 times u_{mf} and soon blows out of the container.

The type A solid on the other hand, has a much larger range of stable fluidization expressed as a multiple of u_{mf}. There is a zone of bed expansion in which no bubbles form and the increased gas velocity all appears as a greater flow of interstitial gas. When bubbling begins, the bed expansion increases slowly. After the terminal velocity is exceeded the bed can still be operated as a turbulent or fast fluidized bed depending on the rate of overheads recycle.

In summary, the particle and gas properties combine with particle size and gas flow to determine the fluidization regime. It is important to be aware that experimental data or correlations derived from one regime may not be applicable to another. Failure



Fluidization Regimes by Velocity (10) Fig. 1-3

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to do so will lead to difficulties in scaling up and scaling down fluidized systems.

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Chapter 2

FIRST CALCULATIONS

Average Particle Size 2.1

Particle size is usually determined by screening to obtain the weight fraction in each sized fraction. There are then several ways of averaging the particle diameter. The one that is most accepted for fluidization calculations is the surface volume diameter, d_{sv}, which is given as:

$$d_p = d_{sv} = \frac{1}{\sum x_i/d_{p_i}} \qquad [2.1-1]$$

where

d_p = average particle size
x_i = weight fraction of average screen size d_{p_i}

This diameter gives greater weight to fine particles and is the method of choice when a particle size distribution is involved (1).

It is universal practice to use the nominal diameter for d_p, whether defined by a screen or an electronic device. This introduces little error since most fluidization processes work on particles that are nearly spherical. However, the surface volume diameter is strictly correct (1). The surface volume diameter is the diameter of a sphere with the same surface area per unit volume as the particle. The volume diameter, d_v is also found in the literature. It is the diameter of a sphere of the same volume as the particle. When only the screen size, d_s, is available for irregular shaped particles, Abrahamsen and Geldart (1) recommend the following relationships:

$$d_v = 1.127 d_s \qquad [2.1-2]$$

$$d_{sv} = 0.871\, d_s \qquad [2.1-3]$$

The most accurate approach is to include a sphericity term to take into account departure from ideality. The term ϕd_s is substituted for d_s

where

ϕ = surface area of a sphere whose volume is equal to that of the particle/surface area of the particle

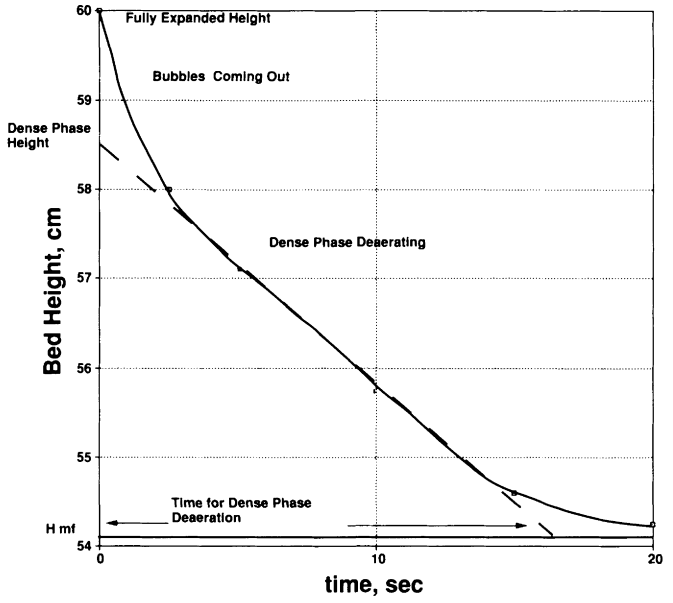
The processes of feedstock variation and attrition usually result in a mixture of particle sizes in a fluidized bed. We recognize that a mixed size bed fluidizes more "smoothly" than a closely sized one. The smaller particles fit between the larger ones and act as ball bearings or a lubricant to make flow easier. A range of particle sizes spanning an order of magnitude is reasonable.

Experimental Evaluation vs. Calculation 2.2

The next several paragraphs discuss calculation of u_{mf}, u_{mb} and ϵ_{mf}. However, if they will be important to one's calculations, it is best to determine these values on a more certain basis by actual experimentation. The minimum fluidization velocity is found by increasing and decreasing flow to the system above and below the minimum fluidization velocity. Pressure drop is plotted vs. velocity and the point at which the increasing pressure and the constant pressure lines intersect is taken as u_{mf}.

Irregularly shaped particles are more difficult to fluidize. A plot of pressure drop vs velocity will a show a peak associated with the extra force needed to unlock the particles. When fluidizing at increasing velocity, additional dp and velocity are required compared to the same measurement made at decreasing velocity. Extremely non–spherical particles such as needles or flakes may not fluidize at all.

The minimum bubbling velocity can be determined visually or by using pressure drop instrumentation to watch for the appearance of fluctuations which correspond to bubbles. Measurement of bed height can give ϵ_{mf} and ϵ_{mb} at the corresponding velocities.



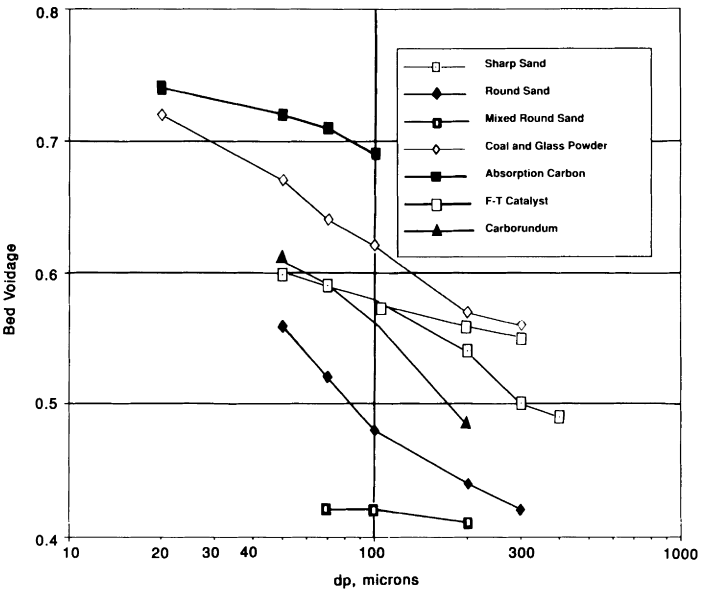
Deaeration Curve for Type A Particles (2) Fig. 2–1
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An experimental procedure which is very informative is the deaeration test (2). The material to be tested is fluidized vigorously and the gas is then instantly turned off. The bed level is recorded as a function of time and a graph like that in shown in Figure 2–1 is obtained. The bed level first drops quickly reflecting the flow of bubbles leaving the bed. An inflection point is then reached as remainder of the gas leaving the bed is that flowing through the interstices. The slope of the second line is u_{mb} and the intercept is h_{mb}. The final bed height can be used to determine ϵ_{mf}.

Minimum Fluidization Velocity 2.3

Traditionally, the expression for minimum fluidization is derived from the Carman Kozeny equation and is given by:

$$u_{mf} = \frac{\epsilon_{mf}^3}{150(1-\epsilon_{mf})} \frac{(\rho_s - \rho_f)(\phi d_p)^2 g}{\mu} \qquad [2.3-1]$$

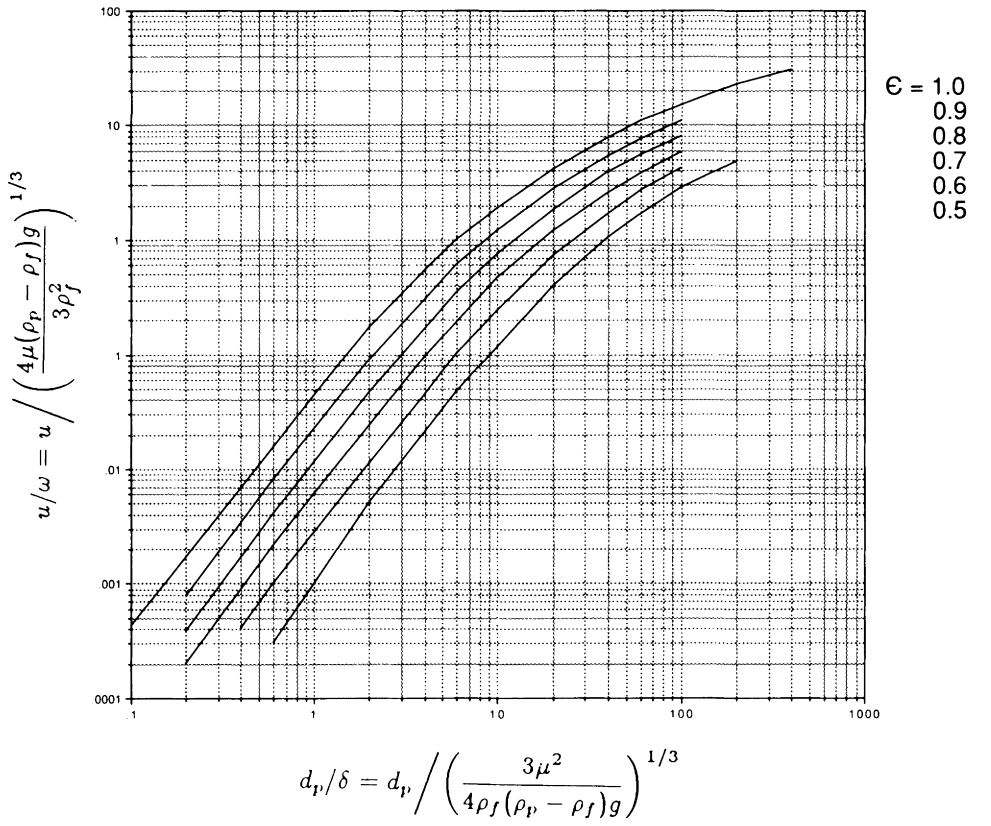


Typical Voidage Data (4) Fig. 2–2

This is the equation for small particles in laminar flow where viscous forces prevail. For larger particles, a term for the inertial effects is added which contains u^2. This requires an iterative solution for the minimum fluidization velocity. More important is that the minimum fluidization velocity is very sensitive to the voidage and we cannot predict the voidage well. Data for ϵ_{mf} are given in Figure 2–2 (3,4). Fine light particles such as catalyst are likely to have minimum fluidization voidage on the order of 0.6 while typical sand and ore particles will be closer to 0.45. Irregular shaped particles tend to pack less well and have a higher ϵ_{mf} than round ones. As particles get finer in the Group A range, ϵ_{mf} also increases. Finally, a wide particle size distribution should give lower voidage since fine particles can be fit into the spaces between larger ones.

An accepted correlation which does not explicitly require the voidage is given by Wen and Yu (5). The original Wen and Yu equation used the volume diameter and is given below.

$$Re_{mf} = (33.7 + 0.0408\ Ga)^{0.5} - 33.7 \qquad [2.3-2]$$



Zenz's Nondimensional Velocity–Diameter Plot Fig. 2–3
redrawn from (7)

where

$$Re_{mf} = u_{mf} d_p \rho_f / \mu$$
$$Ga = d_p^3 \rho_f (\rho_s - \rho_f) g / \mu^2$$

Increasing pressure and increasing temperature are both predicted to decrease the minimum fluidizing velocity and this is confirmed by experiment (6). The particle diameter is obviously a key feature but the effect of the particle size distribution is not given. As mentioned earlier, the surface mean diameter is accepted as being the best to use.

The Wen and Yu equation inherently contains a correlation for the typical minimum fluidization voidage in the systems they correlated. If the voidage at minimum fluidization is known, a better alternative is to use Figure 2–3. Zenz and Othmer (3) and Zenz (7) put the drag coefficient–Reynolds number relationship into a form which can be solved graphically. The particle properties are contained in a diameter term, δ and the gas properties are calculated to a velocity term ω. These are both defined in Figure 2–3.

From the particle properties, δ is calculated and the point d_p/δ is located on the graph at the correct ϵ_{mf}. The term u/ω is read from the graph. And the u_{mf} is obtained after calculating ω from the gas properties.

It is noted that d_p/δ is equivalent to $(4/3Ar)^{1/3}$.

Minimum Bubbling Velocity 2.4

Many equations use the term $(u-u_{mf})$ to determine excess velocity for purposes of determining entrainment parameters or bubble growth. For type B solids, we assume no difference between the minimum bubbling velocity and the minimum fluidization velocity. For type A systems, the minimum bubbling velocity should be used in these equations. The minimum bubbling velocity can be as much as ten times the minimum fluidization velocity. Abrahamsen and Geldart (1) recommend the following in kg–m–s units:

$$u_{mb}/u_{mf} = \frac{2300\rho_f^{0.126}\mu^{0.523}exp(0.716F)}{d_p^{0.8}g^{0.934}(\rho_p-\rho_f)^{0.934}} \qquad [2.4-1]$$

The term F is introduced here as the fraction of solids which are less than 45 microns. This is the same as a 325 mesh screen which is the smallest screen in common commercial use. Many commercial enterprises define fines as less than 45 micron material.

It is again noted that typical industrial processes operate so far above u_{mf} that precise determination of the datum is not needed. In situations where bubble size and growth are calculated, the minimum bubbling velocity is of genuine interest. If these quantities are critical to modeling, laboratory measurements should be made.

Terminal Velocity 2.5

For applications where we need to determine terminal velocity of the particles, the material is usually fine and in the region where Stokes law applies. Thus for particle Reynolds numbers less 0.4:

$$u_t = d_p^2(\rho_p - \rho_f)g/18\mu \qquad [2.5-1]$$

Terminal velocity can also be calculated from Figure 2–3 for any particle size. The properties of the solid and the gas are used to calculate d_p/δ and ω. The graph is used to find u/ω at $\epsilon = 1$ and d_p/δ; the terminal velocity is then u/ω times ω.

Bed Expansion 2.6

As bubbles fill the bed and occupy space, the bed level must rise. The degree of bed expansion will be a function of the bubble rise velocity u_{mb}, and of the interstitial gas velocity u_ϵ. Additionally, the bed expansion will be a function of the fluidization regime and the operating equipment. In this discussion we will not be discussing slugging beds and are assuming that we are talking about freely bubbling equipment.

If we continue the use of Figure 2–3, the bed expansion can be estimated for a bed operating below the terminal velocity of the largest particles as per Zenz (7). The minimum fluidization velocity is first calculated or determined experimentally. The minimum fluidization condition can then be found on the plot at the actual operating condition of bed size consist and gas operating environment. The largest particle size in the bed is then used to determine d_p/δ at $\epsilon = 1.0$ on the plot. At this latter condition, the largest particle in the bed would be elutriated

and the bed voidage will be zero. A line is drawn between the conditons of minimum fluidization and complete entrainment. Bed voidage and therefore expansion can then be determined at all velocities by reading the points off the tie line.

Interestingly, many commercial fluidized beds operate above the terminal velocity of the largest particles. These systems are turbulent and fast fluidized beds. There is no definite bed level to use for an expansion calculation. However, there are some correlations available for estimating the bed density and how varying the operating conditions changes the observed density.

For a turbulent bed, the density-velocity relationship is based upon the Richardson–Zaki equation (8).

$$u/u_t = \epsilon^n \qquad [2.6-1]$$

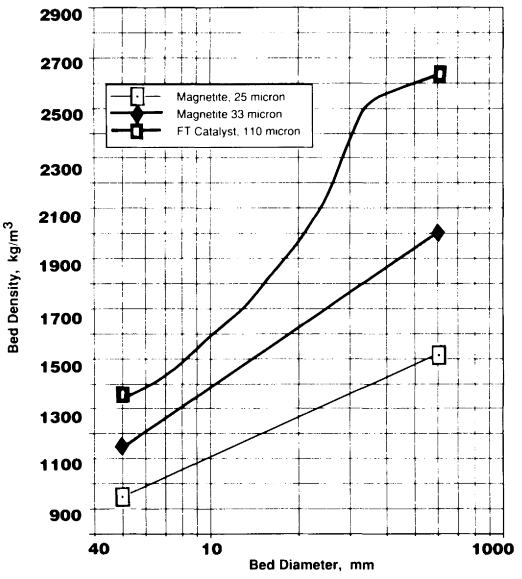
The above equation with $n = 4.6$ describes expansion in liquid fluidized systems. Avidan (9) examined turbulent fluidization and found that, in order fit data for cracking catalyst, u_t needed to be increased enormously to the order of several meters per second and was renamed u_t^*. The exponent, n, varied from 4 to 14 with different materials.

A short cut method based on extensive experience with catalytic cracking units is given by King (10) as:

$$\epsilon = \frac{u+1}{u+2}$$

with u given in m/s.

The above discussion shows how bed expansion can be altered as the regime of fluidization is altered by changing velocity. Another way that density changes is by vessel size. Squires et al (11) quote data as in Figure 2–4 which show that measured bed density can increase as the reactor size is increased. In small, short reactors, the gas is well distributed and bubble growth is not extensive. As diameter and height are increased, the gas bubbles tend to shift toward the center as they rise. The large bubbles rise quickly and occupy less volume so that the overall density increases. Obviously there is a considerable radial density gradient in such a system.



Density as a Function of Bed Diameter Fig. 2–4
redrawn from (11)

A Second Look at the Minimum Bubbling Condition 2.7

Gibilaro et al (12) have published a correlation for the voidage at the minimum bubbling condition which takes into account the effect of temperature and pressure. Bubbling begins when the equation below is true.

$$\left(\frac{g d_p(\rho_p - \rho_f)}{u_t^2 \rho_p}\right)^{0.5} - 0.56n\,(1-\epsilon)^{0.5}\epsilon^{n-1} = 0 \qquad [2.7-1]$$

where

n = the Richardson–Zaki exponent for expansion

If we consider the material to behave as a typical fine powder we can let $n = 4.8$ and $u_t = (\rho_p - \rho_f)g d_p^2/18\mu$. Then the criterion reduces to:

$$\frac{\mu}{\left((\rho_p - \rho_f)\rho_p g\ d_p^3\right)^{0.5}} - 0.149\,(1-\epsilon)^{0.5}\epsilon^{3.8} = 0 \qquad [2.7-2]$$

The correlation matches the observations that bubbling is reduced as temperature or pressure are increased or particle size is decreased. If experimental data are available for n or u_t, the experimental value should be used.

Bed Pressure Drop 2.8

The bed pressure drop is equal to the weight of the bed. This gives:

$$dp = (1 - \epsilon)\rho_s g h \qquad [2.8 - 1]$$

where

dp = pressure drop in Pa for S.I. units

g is replaced by g/g_c for English units to give pressure drop in units of lb/ft^2

This is the most reliable calculation. When a close match between the bed weight and the measured pressure is not obtained, then look for a physical cause for bypassing and check the pressure measurement.

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Chapter 3

GAS DISTRIBUTOR DESIGN

Functions of the Grid 3.1

The gas must be distributed uniformly into the fluidized bed without causing attrition of the bed material. There must be enough holes to ensure local uniformity of aeration as well as uniformity based on the total cross sectional area. Eventually, a point is reached at which adding more holes is no longer cost effective. In most beds, the distributor also is the floor which supports the weight of the bed.

Grid Pressure Drop 3.2

Due to the bubbling action of the fluidized bed, there are constant small changes in the local pressure drop at the grid. Gas inevitably tries to enter the bed in the zone of lowest pressure drop. The distributor pressure drop therefore has to be large enough to overcome the small local pressure disturbances of the bed. If the pressure drop is too small, gas will end up flowing through only some of the openings and establishing flow paths with high voidage and low pressure drop. At the same time, other holes would have negligible flow and remain"closed" or nonbubbling.

When the fluid bed is at rest, particles will partially block some of the holes. The initial flow of gas must blow out the blocking particles in order for free flow to develop. Some of the holes will open before others. Once they are open, the gas will preferentially flow through the least blocked openings and the possibility exists that flow will never develop in the blocked holes. The pressure drop has to be large enough to open all the holes at normal operating conditions if the bed is to operate properly.

The above criteria key the design of the grid. In practice, the required pressure drop is set by operating experience. For upflow, the design pressure drop should be at least 0.3 of the bed pressure drop, *dp*, at minimum expected gas flow and maximum expected bed weight. If pressure drop is not a cost factor, 0.4 or 0.5 is a more comfortable factor.

Can one design for too much pressure drop? Yes. A comfortable margin of safety is to design the grid *dp* for 100% of the bed *dp*. This is often no strain for systems in which the gas is coming from a relatively high pressure source. The only problem is that, while the gas will be well distributed, the velocity might become high enough to cause attrition problems. The issue is discussed later in this chapter and in the chapter on attrition.

Downflow jets do not suffer the problem of having to clear the holes and it is generally accepted that a lower pressure drop can be used. However, it becomes essential that the manifold be designed so that the gas is well distributed from the point of view of pressure drop in the piping. For this kind of system a pressure drop of 0.2 of the bed weight is suitable for coarse materials. Fine material on the order of less than 75 microns can be sucked into the distributor piping and cause erosion as it exits. In such cases, the pressure drop should still be 0.3 of the bed weight. (1)

The pressure drop can be related to the orifice gas velocity by the equation,

$$u_o = 0.8\left(\frac{2g\ dp}{\rho_f}\right)^{0.5} \qquad [3.2-1]$$

where

u_o = the velocity in the grid orifice

and the orifice coefficient for blowing into a fluid bed is taken as 0.8 according to Zenz (1). The gas conditions in the hole are taken as those in the plenum.

Hole Size and Number 3.3

Once the pressure drop and hole velocity are set, the total hole area is obtained from,

$$u_o = Q/(A_o N_o) \quad \text{and} \quad AoNo = Q/u_o \qquad [3.3-1]$$

where

Q = the volumetric flow rate in the holes m^3/s
N_o = the number of holes

If the holes are on a triangular pitch,p, the number of holes per unit grid area is $2/(\sqrt{3}\ p^2)$ and for a square pitch, $1/p^2$ (2). Once the velocity is determined, the hole diameter and pitch are related by allowing (for triangular pitch) the hole area per unit area, $A_o 2/(\sqrt{3}\ p^2)$ to be equal to the total hole area determined above, Q/u_o, divided by the total grid area. Then,

$$\frac{\pi}{4} d_o^2 \frac{2}{\sqrt{3}\ p^2} = \frac{Q}{u_o} \frac{4}{\pi d_t^2} \qquad [3.3-2]$$

This leads to the relationship between p and d_o:

$$p = \left(\frac{u_o}{Q} \frac{\pi^2 d_t^2}{8\sqrt{3}} \right)^{0.5} d_o \qquad [3.3-3]$$

We now have determined the hole size as a function of the pitch for a desired pressure drop. The next question is whether the holes should be made as small as possible in order to have small bubbles which will presumably provide for smoother, more effective fluidization. If the jet length is calculated as given in the next section, the initial bubble size can estimated as one half the jet length. Zenz (3) reports that bubble merger occurs between rising bubbles if they are closer than $1.5\, d_b$. Thus the designer may not necessarily achieve smaller bubbles by having more holes spaced closer together. It is recommended that the possibility of bubble merger be checked along with a calculation of jet length on all designs.

Jet Length 3.4

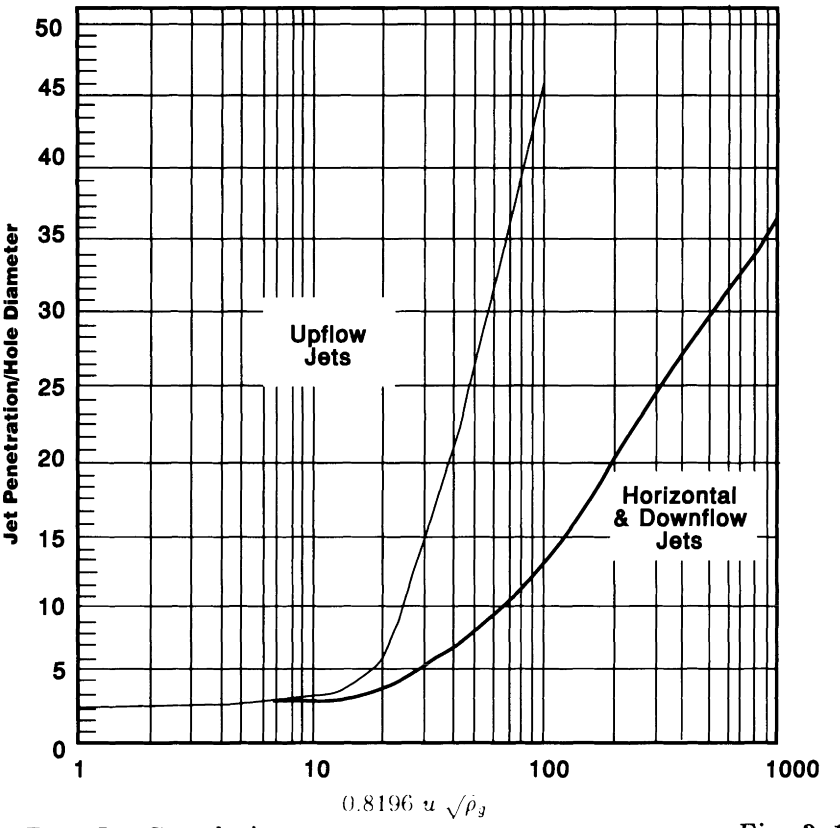
At low velocity, gas issues from the grid in a series of bubbles. Rowe et al (4) report that these bubbles are usually at the rate of 8 per second. In order to move particles away from the hole so that a bubble can form, gas must travel past the bed particles to exert the required drag force. Rowe (4) found that the volume of the initial bubbles is typically 1/2 of the gas entering the bed. Gas then reenters the bubbles as they rise away from the distributor.

In commercial units, gas emanating from the grid hole will typically have sufficient velocity to create a jet above the hole. Correlations of jet length have been based on observations of industrial units in which the jets caused erosion of internals and based on laboratory results. Unfortunately, existing correlations can differ by an order of magnitude (5). It is suggested that two correlations be checked for estimating the jet length. The correlations of Merry (6,7) are given below for upflow and downflow or horizontal flow. The correlations are different because it is easier for a jet to penetrate upwards than in the horizontal or vertical direction. Presumably, the particles can move out of the way more easily when they are not constrained by a wall or a grid plate. For upflow jets (6),

$$l_j/d_o = 5.2\left(\frac{\rho_f d_o}{\rho_p d_p}\right)^{0.3}\left[1.3(\frac{u_o^2}{g d_o})^{0.2} - 1\right] \qquad [3.4-1]$$

For horizontal or downflow jets (7),

$$l_j/d_o + 4.5 = 5.25\left(\frac{\rho_f u_o^2}{(1-\epsilon)\rho_p g\ d_p}\right)^{0.4}\left(\frac{\rho_f}{\rho_p}\right)^{0.2}\left(\frac{d_p}{d_o}\right)^{0.2} \qquad [3.4-2]$$

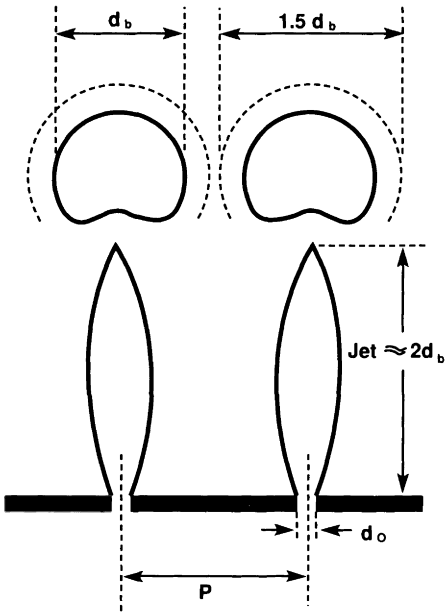


Zenz Jet Correlation Fig. 3–1
redrawn from (8)

where

l_j = length of the jet

Another approach is the correlation of Zenz (8) for both upward and downflowing jets. The correlation is given as Figure 3.1. The Zenz correlation was developed from data gathered on large scale catalytic cracking units. It is reported that the correlation is also applicable to gas solids suspensions if the density of the mixed suspension is substituted for the gas density. Since particle density and diameter terms are absent, the correlation should be used with care if it is applied to materials which differ markedly from cracking catalyst.



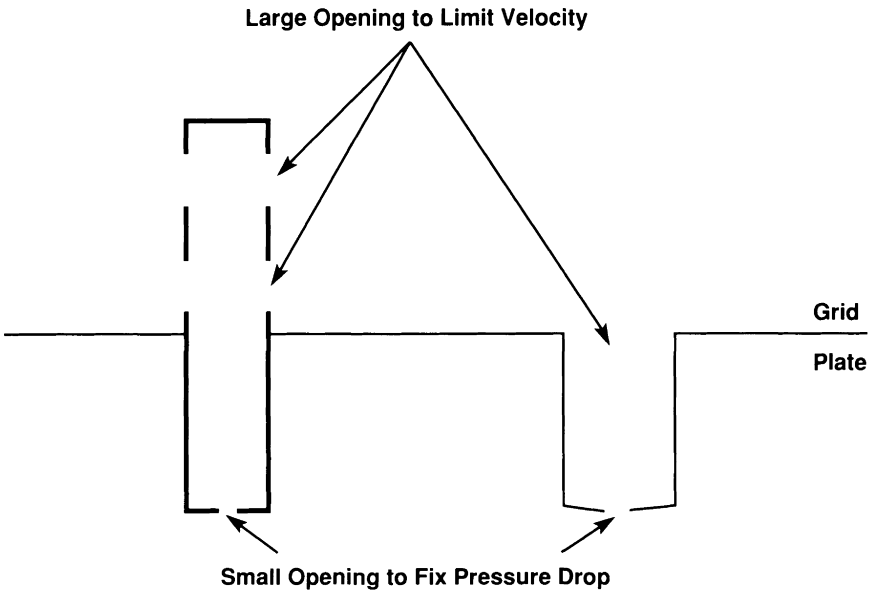
Jets and Bubble Merger Above the Grid Fig. 3–2
redrawn from (3)

Figure 3.2 sketches what a bubbling jet looks like and how the bubble cloud extends past the nominal bubble facilitating early bubble merger just above the grid.

Jets lose their integrity by leakage of gas into the solids and by expansion. Placement of a jet next to a wall therefore tends to stabilize it. Correlations based on experiments in which the wall can interfere with the jet are expected to yield jet penetration values which are high.

Particle Attrition 3.5

High velocity in the grid hole can lead to a jet which causes attrition of bed material. The degree of concern that this raises is a function of the material being fluidized. Velocities under 30 m/s are generally safe while Geldart (2) reports that velocities over 90 m/s generally are not. It would be best to do laboratory studies on the material of interest in any case. The attrition rate increases approximately according the 2.5 power of the gas



Distributors to Control Attrition Fig. 3–3
redrawn from (2)

velocity. Grid attrition is discussed as a separate topic in a later chapter.

Figure 3.3 shows two orifice designs which can be used to lessen attrition. In the first, most of the pressure drop is taken in the small diameter pipe so that the hole area can be enlarged to lessen the velocity of gas entering the bed. In the second, the pressure drop is taken in a small orifice and the gas is expanded into a shroud to reduce the velocity seen by bed material. The large diameter shroud has to be long enough for the expanding jet velocity profile to fill the pipe. A shroud that is too short will allow solids to fall into the shroud and be entrained in the jet causing severe erosion problems. Zenz (8) recommends that the length of the shroud be at least,

$$l_{min} = (d_{shroud} - d_o)/(2\ tan\ 5.5°)$$

or

$$l_{min} = (d_{shroud} - d_o)/0.193 \qquad [3.5-1]$$

Grid Variations 3.6

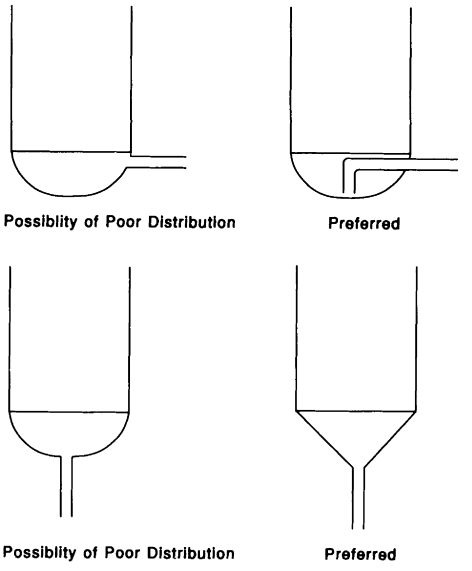
There are may grid design variations that can be envisioned. Geldart (2), Kunii and Levenspiel (9) and Zenz (8) display a number of these and comment on the origins of the designs. It is up to the designer to make a selection based on the needs of his system. All else being equal, a simple perforated plate is preferred; the above references give several examples where complex designs gave harmful flow patterns for gas and solids.

Grid Seals 3.7

A common problem in industrial beds is that the grid has not been properly sealed to the walls. This allows gas to bypass the grid plate and fluidization is lost or becomes uneven. The grid seal design is a mechanical issue which can easily be overlooked in process design. A good design is to weld the grid plate to the vessel wall and to build in a mild concave shape so that expansion can be accommodated as the vessel is heated and cooled. Another technique is to use gasketing above and below while leaving room to move horizontally as thermal expansion and contraction takes place.

The greatest difficulty occurs in cooling and heating cycles which place the bed and plenum at different temperatures. This may occur during a heatup cycle where hot gas is in the plenum. On cooldown, we may use cool gas in the plenum while the bed is hot. If all these conditions are not accounted for, the grid plate can warp enough to lose the seal and thereafter ability to fluidize the bed. If both refractory and metal parts are present, these will expand and contract to different degrees which may cause mechanical problems.

Grid warpage was a significant problem in early fluid bed combustors. The units operated well in startup and operation. However, on shutdown, the hot bed defluidized on the plate while the still air in the plenum provided no cooling. Grid warpage was a common problem. A typical solution is to use fluidizing nozzles which extend about ten centimeters above the grid plate. An insulating layer of defluidized bed solids then insulates the grid plate from the bed.



Some Possible Plenum Arrangements Fig. 3–4

Plenum Design 3.8

The purpose of having a significant grid *dp* is to distribute gas from the plenum. While *dp* variation associated with the bed is the dominant consideration, the design of the plenum should be arranged to make the grid distribution as easy as possible. Inertia causes gas streamlines to continue in the direction they were going. A pipe entering the plenum below the grid should be positioned to enhance gas distribution as much as possible. Figure 3.4 shows several alternatives with a preferred and less preferred orientations. Gas distribution in the plenum is usually not a problem but it is sensible to pay attention to this detail as a preventive action. Recent literature is still discussing the importance of plenum orientation (10).

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Chapter 4

BUBBLE GROWTH AND STABILITY

Overview 4.1

The nature of bubbling in the bed is determined by the solids, the gas velocity and the scale of the bed. High velocity limits bubbling when there is so much gas that it becomes the continuous phase and the solids are the discontinuous phase. We then see mostly gas with streamers of solids. This condition is considered the onset of turbulent fluidization. Fast fluidization occurs at even greater velocity and voidage. Fluid cracking units and most commercial catalytic operations operate at high velocity and therefore do not contain bubbles.

In terms of Geldart solids, type C, cohesive materials are difficult to fluidize and the gas tends to make channels in the solid and escape through those channels without aerating most of the bed. Bubbling is difficult to achieve. At high enough gas throughputs and with good distribution, it may be possible to aerate the whole mass. However, rather than obtaining discrete bubbles, we see a continuous gas phase containing well dispersed solids.

Group A solids are characterized by not bubbling until a velocity, u_{mb}, well above the minimum fluidizing velocity is reached. This can be several fold u_{mf} (1). Bubbles then grow as they rise, but they eventually reach a maximum size which is a function of the particle terminal velocity.

Group B solids start to bubble once u_{mf} is exceeded and the bubbles can grow to enormous size limited only by the container.

Group D solids are very coarse and tend to produce jets rather than bubbles.

The container can have a profound effect on bubbles. In lab units and in pilot plants, it is possible for bubbles to expand to the point where they fill the container and slugging begins. This is an equipment based limitation to bubble size and can cause severe scaleup problems if it is not recognized.

Bubble Growth 4.2

Bubbles tend to coalesce and grow as they travel up the bed. This mechanism has the capability of producing bubble diameters on the order of meters in large reactors. It is suggested that the expression of Darton et al (2) be used to calculate bubble size as a function of height.

$$d_b = \frac{0.54(u - u_{mb})^{0.4}(h + 4\sqrt{A_t/N_o})^{0.8}}{g^{0.2}} \qquad [4.2-1]$$

where

h = height in the bed

A_t/N_o = grid area per hole.

For a porous distributor, the grid area per hole is equal to zero.

Slugging 4.3

When the bubble reaches 50-75% of the vessel diameter, the system is said to be slugging. In small containers, solids can be lifted as plug; in larger units they can rain along the sides of the container as the bubble rises up the center. The problem with slugging is associated with scale up. When slugging occurs, bubble size and velocity are limited and mass transfer is much better than would be observed in a larger diameter vessel with greater bubble size. Failure to be aware of this effect can lead to scale up problems.

Maximum Stable Bubble Size 4.4

If one used the Darton correlation to calculate the bubble size for an industrial system such as catalytic cracking, the bubble size turns out to be on the order of meters. Such a bubble would violently shake the unit as the tons of catalyst are splashed when it surfaced. In addition, mass transfer from such an enormous bubble would be so poor that it would limit the ability of reaction to take place. As neither of these phenomena take place, there must be a mechanism operating which limits the maximum stable bubble size. Davidson and Harrison (3) first postulated a mechanism for maximum stable bubble size. However, the sizes predicted are much smaller than those actually observed.

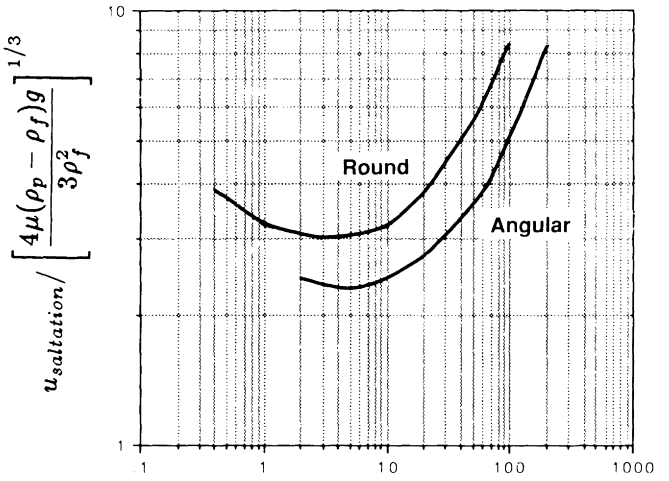
Matsen (4)and Werther (5) investigated maximum stable bubble size and found that bubble size decreased as the level of fines increased.

Horio and Nonaka (6) experimented with a bed into which they injected large bubbles and then measured the breakup into smaller ones. Although some of the parameters in their model remain to be further refined, they present a model in which bubble size is a function of splitting and coalescence frequencies. The equilibrium bubble is one which is constantly splitting and coalescing.

The particles in the bubble roof are prevented from falling into the bubble by the flow of gas circulating upwards through it. The bubble is split when solids at the bottom are swept into it. A correlation for stable bubble size is expected to use the particle terminal velocity as a parameter. Geldart (7) suggests that the maximum stable bubble size be determined from the bubble rise equation where the bubble velocity is taken as u_t based on 2.7 d_p. So that

$$u_b = u_t \text{ of } 2.7d_p = 0.71\sqrt{g\ d_b} \qquad [4.4-1]$$

It is claimed that bubble sizes calculated in this way approximately match the bubble size back calculated from observed bed density in large units.



Saltation Velocity per Zenz (8) Fig. 4–1
Redrawn from (8)

Zenz suggests that the maximum stable bubble size is related to the saltation velocity based on the mean particle size. Saltation velocity is defined as the point at which solids start to fall down in a horizontal dilute phase conveying system. The saltation velocity is obtained from Figure 4.1 using the curve for angular particles. After obtaining the saltation velocity,

$$u_b = u_{saltation} = 0.71\sqrt{g\ d_b} \qquad [4.4-2]$$

This correlation also gives good results in industrial systems.

In a test calculation for an 80 micron particle size with a density of 2800 kg/m^3 fluidized with air, Geldart's maximum size was 54 cm vs. Zenz's size of 56 cm. Both results are reasonable and are probably as accurate as can be confirmed with our existing knowledge base. The two correlations differ greatly as particle size decreases toward the cohesive range. Extensive data is difficult to gather because maximum stable bubble size experiments need large units with solids recovery systems. A satisfactory and well accepted method for this calculation is still required but the effect of fines is universally accepted. The effect of velocity is less certain although once turbulent fluidization is achieved, the gas phase becomes continuous and the concept of bubbles becomes quite unclear.

Effect of Bubble Size on Bed Expansion 4.5

Due to bubble growth, the observed bed expansion will be a function of vessel size. In a small pilot or lab facility bubbles will not grow to very large size. However, as the system is scaled up, bubble size increases and with it bubble velocity according to the relations just given. As bubble velocity increases, the fraction of bubble volume in the bed must decrease. Accordingly, the overall bed expansion at constant fluidization parameters must decrease as bed height increases. Furthermore one would expect density to increase as a function of height along the bed so that a local measurement of bed density would not apply to the entire bed.

The bubble volume–velocity relationship affords another check on the actual bubble size in the bed. The bed density is the particle density reduced by voidage at minimum bubbling and by the bubble voidage.

$$\rho_{bed} = (1 - \epsilon_{mb} - \epsilon_b)\rho_p \qquad [4.5-1]$$

The gas volume passing through the bed as bubbles is either the bubble velocity times the bubble area or the total gas less the interstitial gas velocity times the total area:

$$Q_{bubble\ gas} = (u - u_{mb})A_T = u_b A_b \qquad [4.5-2]$$

$$\epsilon_b = A_b/A_T = (u - u_{mb})/u_b \qquad [4.5-3]$$

The above logic is flawed by the inherent assumption of gas continuity between bubbles but it shows how a voidage measurement can give a check on bed density vs bubble size.

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Chapter 5

MIXING AND SEGREGATION

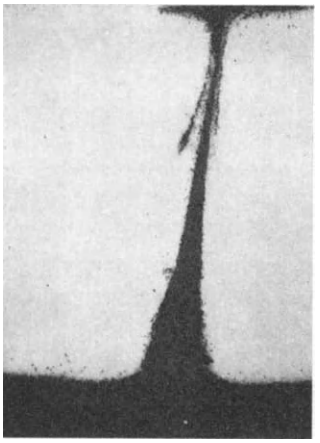
Solids Mixing 5.1

Solids in a fluid bed are usually considered to be fully mixed. And most industrial fluidized beds are operated at velocities high enough so that they are indeed fully mixed. However, there are times when it is of specific interest to determine how long it takes to mix material into the bed from an unmixed condition. Alternatively, there may be a heat transfer concern where it is desired to estimate whether there will be a temperature profile in the bed. For example we may be burning a fuel at the bottom of the bed and want to be sure that there will be a minimal temperature gradient in the system. We may be passing solids through a heat transfer system and want to be certain that there is no temperature gradient. In some cases, the bed contains solids of differing size or density and there is uncertainty as to how well mixed they are. Finally, in large units feed may be introduced in the side of the vessel and we want to be reasonably certain that the solid will be evenly distributed in the bed.

Mechanism of Vertical Mixing 5.2

The mixing action in a fluidized bed is driven by rising gas. Rowe and Partridge (1,2,3) graphically showed that as a bubble rises through the bed it brings a wake of material with it and that its passing also pulls along a trail of upwardly transported material called the drift. Their findings are reproduced in Figure 5.1.

The experiments were carried out in a fluidized bed of two layers containing different color materials. Sectioning the bed after passage of a bubble gives clear evidence of the mechanism. A bubble carries a portion of the lower bed with it behind the bubble in a section called the wake. As it travels up the bed, the top of the bubble pushes solids up and to the side; these then move down the sides and leave a trail of material that has been

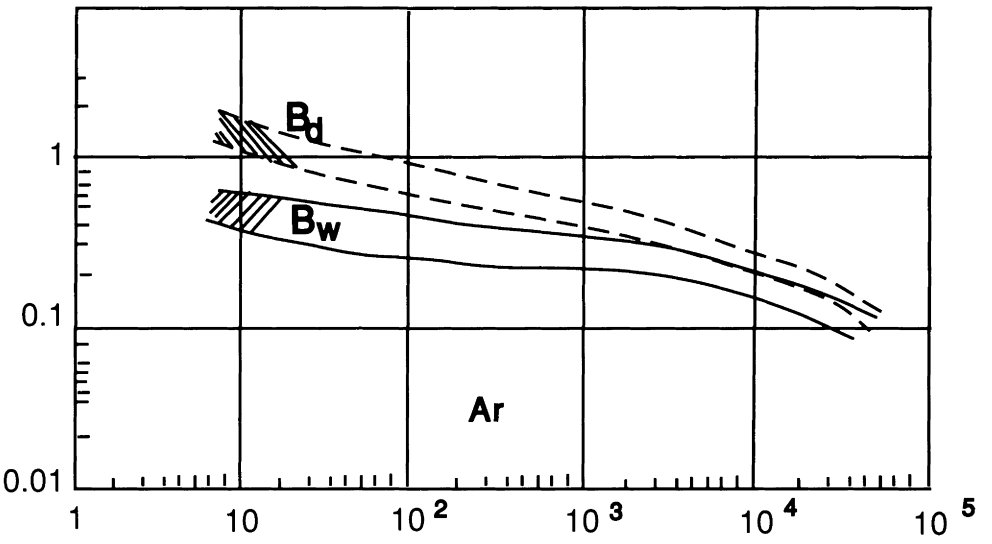


Mechanism of Bubble Induced Mixing Fig. 5–1
reprinted from (2) by permission of Academic Press

pushed up from below. As the bubble rises, it pulls material up from below to fill the space the bubble displaced on its upward journey through the bed. When the bubble breaks the surface, the nose or front of the bubble lifts some bed contents up out of the bed. If these solids are not entrained, they fall back onto the surface. Similarly the wake is traveling with the bubble and may leave the bed surface too. The drift is also pulled toward the surface albeit at a lower velocity. Therefore the wake and drift end up on the surface of the bed thereby producing the intense top to bottom mixing mechanism. By material balance, the surface material must move downward to replace the upflowing solids.

Baeyens and Geldart (4) correlated the wake and drift fractions in terms of fractions of bubble volume, Q_b, as defined below versus the Archimedes or Galileo number as shown in Figure 5.2.

$$Q_b = A_t(u - u_{mb}) \qquad [5.2 - 1]$$



Wake and Drift Fractions Fig. 5–2

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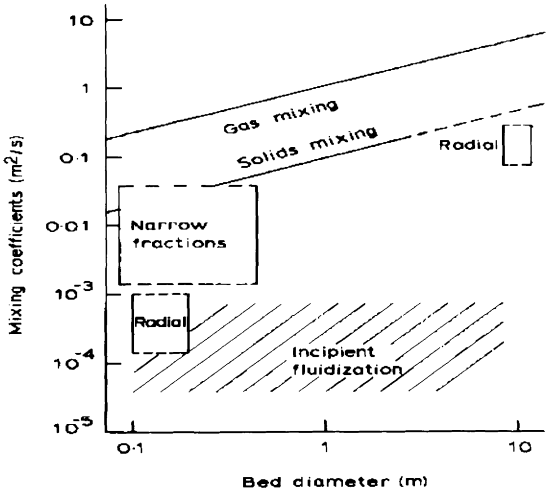
The wake travels upward with the velocity of the bubble while the drift travels upward at about 0.38 the velocity of the bubble. Therefore, the upward solids flux can be represented as:

$$Q_s = Q_b(B_w + 0.38B_d) \qquad [5.2-2]$$

where

Q_s = the solids volumetric mixing rate at ϵ_{mb}
B_w = the bubble wake fraction
B_d = the bubble drift fraction

The above expression can be used as the mixing rate of the fluidized bed.



Axial Solids Backmixing vs Diameter Fig. 5–3
from (7) by permission of Academic Press

Radial Mixing 5.3

The mechanism discussed in the previous section thoroughly mixes the bed from top to bottom. It is generally recognized that the radial mixing is about 10% of the vertical mixing. Radial mixing is important to systems in which solids are fed to the bed below the bed level. Zenz (5) gives the first consideration that the feed system should not be backed up by inability of the bed to remove new solids from the feed point. He recommends that the solids feed rate be less than the calculated bed solids circulation rate.

Definitive references to describe radial mixing in commercial systems are limited. But design of the number of feed points in a bed depends on being able to distribute the solids into the system with assurance that they will not all be reacting at the feed point.

Diffusion Model for Solids and Gas Backmixing 5.4

Mixing can be evaluated via a diffusivity term which is applied to solid or fluid phases in a manner analogous to single phase flow systems. Levenspiel (6) gives a clear discussion of the governing equations and how they are applied. We recognize that the mechanisms causing fluid bed backmixing are different but the model is convenient to use and gives a good fit to observed bed behavior.

Van Deemter (7) gives information on solids and gas diffusivities which are presented in Table 5.1 and in Figure 5.3. Figure 5.3 uses data obtained from backmixing tracer tests with non-adsorbing gases. If the gas is adsorbed on the solid surface, the backmixing will be greatly enhanced.

Range of Solids Eddy Diffusivities *Table 5–1*

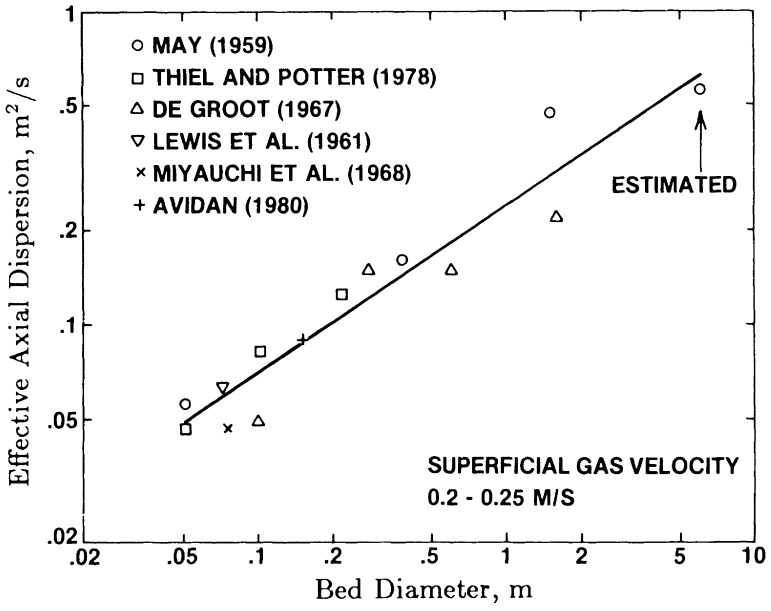
Vessel Diameter, m	E_s, m^2/s
0.1	0.002–0.03
0.3	0.01–0.1
0.6	0.05–0.2
1.5	0.2–0.5
3.0	0.3–1.0

from J.J. Van Deemter (7) by permission of Academic Press

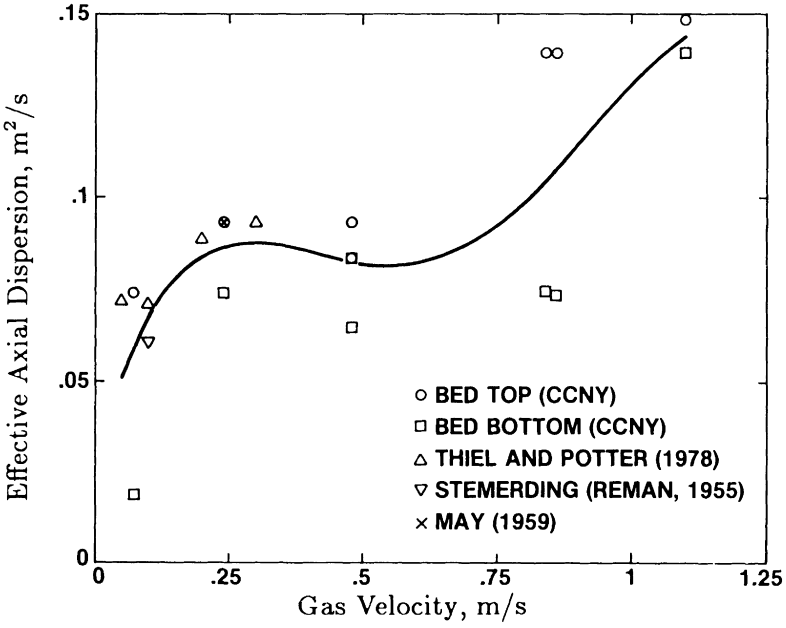
Data are quite scattered and somewhat sparse. The backmixing is affected by particle size; increasing the fines content reduces solids and gas backmixing in large vessels. At conditions close to minimum fluidization, the lack of bubbles greatly reduces mixing.

Avidan (8) studied gas and solids backmixing in fast and turbulent fluidized beds of catalyst. Figure 5.4 shows the effect of diameter on the effective axial dispersion coefficient similar to that given by van Deemter (7). On a log–log scale, axial diffusion is a straight line function of diameter. Figure 5.5 shows the effect of velocity on solids axial diffusion coefficient in a 15 cm unit. The dispersion coefficient showed an increasing trend with velocity up to 1.1 m/s. The figures display the effective axial dispersion coefficient which is used in the dispersion equation. This is related to the apparent dispersion coefficient by:

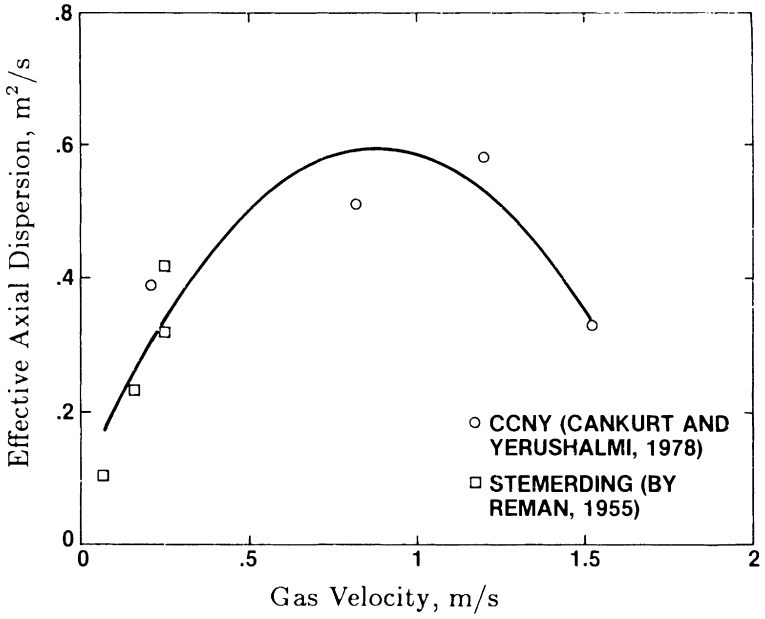
$$D_{apparent} = (1 - \epsilon) D_{effective} \qquad [5.4 - 1]$$



Solids and Gas Eddy Diffusivity Fig. 5–4



Axial Solids Backmixing vs Velocity Fig. 5–5
from (8) with permission

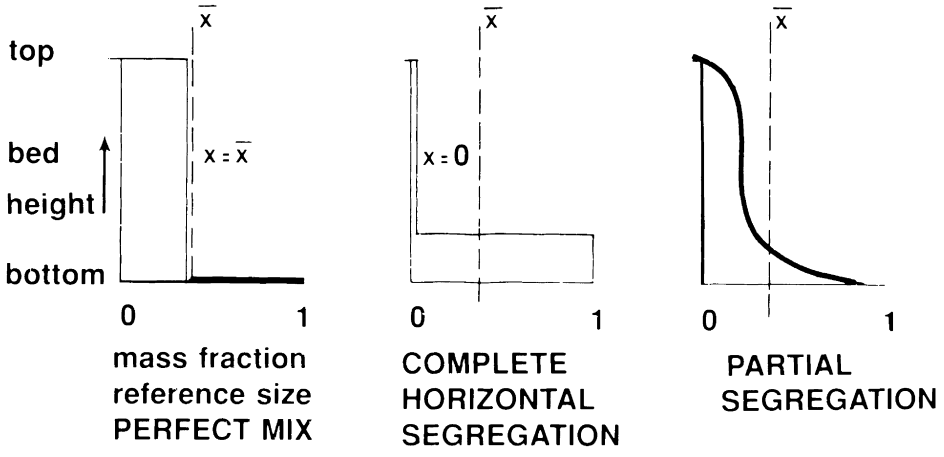


Axial Gas Backmixing Fig. 5–6
from (8) with permission

The effective gas axial diffusion coefficient, Figure 5.6, shows an increasing value with velocity till about 0.8 m/s and then decreases with velocity. The backmixing peaks in the turbulent regime and then falls as "fast" fluidization begins. The effect of diameter was not studied, but it is expected that it would be similar to the diameter effect on solids diffusion.

Segregation 5.5

At high velocities, the fluid bed will mix large and fine particles and dense and light particles. As velocity is decreased, large or dense particles may sink to the bottom and defluidize. In the fluidization literature, the particles which tend to move to the bottom of the bed are called "jetsam" and those that form the top layer are called "flotsam" (9). In this work we shall call the particles in the top layer floaters and those in the bottom layer sinkers.



Mixing Index Definition Fig. 5–7

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Segregation can be quantified via the mixing index as shown in Figure 5.7. The fraction of sinkers in the whole bed is $\bar{x}$ and the fraction in the top layer is x. When the bed is completely mixed, the top layer contains the same heavy content as the overall bed and the mixing index, $M = x/\bar{x}$, is 1. When the bed is completely segregated, there are no sinkers in the top layer and the mixing index is zero.

The mixing index is a function of velocity and the fluidization characteristics of the dissimilar solids. It is correlated using the following equations (9):

$$M = x/\bar{x} = \frac{1}{1 + e^{-z}} \qquad [5.5 - 1]$$

Where z is the velocity ratio defined by,

$$z = \frac{u - u_{TO}}{u - u_{mf\,floaters}} \exp(u/u_{TO}) \qquad [5.5 - 2]$$

We define u_{TO}, the takeover velocity, as the velocity at which the mixing index is 0.5. The takeover velocity is given by:

$$\frac{u_{TO}}{u_{mffloaters}} = \left(\frac{u_{mfsinkers}}{u_{mffloaters}}\right)^{1.2} + 0.9\left(\frac{\rho_H}{\rho_L}\right)^{1.1}\left(\frac{d_H}{d_L}\right)^{0.7} - 2.2\sqrt{\bar{x}}\left(1 - \exp(\frac{-h}{d_t})\right)^{1.4} \qquad [5.5-3]$$

where

ρ_H and d_H = the density and diameter of the heavy particles
ρ_L and d_L = the density and diameter of the light particles

Depending on magnitudes involved, a system with large light particles and small dense particles could have either one sink. The u_{mf} ratio is defined by the larger u_{mf} belonging to the sinkers.

A way of looking at segregation of disparately sized material of the same density is described by Chiba and Nienow (10). At low velocity, the large particles form a layer on the bottom. Increasing the velocity gets one to u_{cf}, the velocity at which the system is completely fluidized. At this point, the bottom layer no longer distinctly exists. They give the equation,

$$1 - \frac{u_{cf}}{u_{mfsinkers}} = \left(\frac{1}{(u_{mfsinkers}/u_{mffloaters})^{1-x^2}} - 1\right)^2 \qquad [5.5-4]$$

References **5.6**

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Chapter 6

HEAT TRANSFER

Introduction 6.1

One of the reasons fluid beds have such wide application is their excellent heat transfer characteristics. Particles entering the fluidized bed rapidly reach the bed temperature. Particles within the bed are uniformly at the same temperature. Gas entering the bed reaches the bed temperature within a few centimeters. Finally, heat transfer to surfaces for heating and cooling is excellent.

Particle to Gas Heat Transfer 6.2

Heat transfer between gas and particles is so good because of the enormous particle surface area available. A fine particle in a fluid bed can be considered to have a uniform internal temperature. An exception may occur if an extremely energetic process, such as burning coal, is taking place.

The lowest heat transfer coefficient to be discussed is from a particle to a still fluid and is given by the Nusselt number, Nu, equal to two.

$$Nu = \frac{h_c d_p}{k_f} = 2 \qquad [6.2-1]$$

where

h_c = the convection heat transfer coefficient
k_f = thermal conductivity of the fluid

For heat transfer in a moving fluid medium, Zenz (1) gives the expression:

$$Nu = 2.0 + 1.3Pr^{0.15} + 0.66Pr^{0.31}Re_p^{0.5} \qquad [6.2-2]$$

where

Pr = Prandtl number based gas properties, $c_p\mu/k_f$
Re_p = particle Reynolds number, $d_p\rho_f u/\mu$

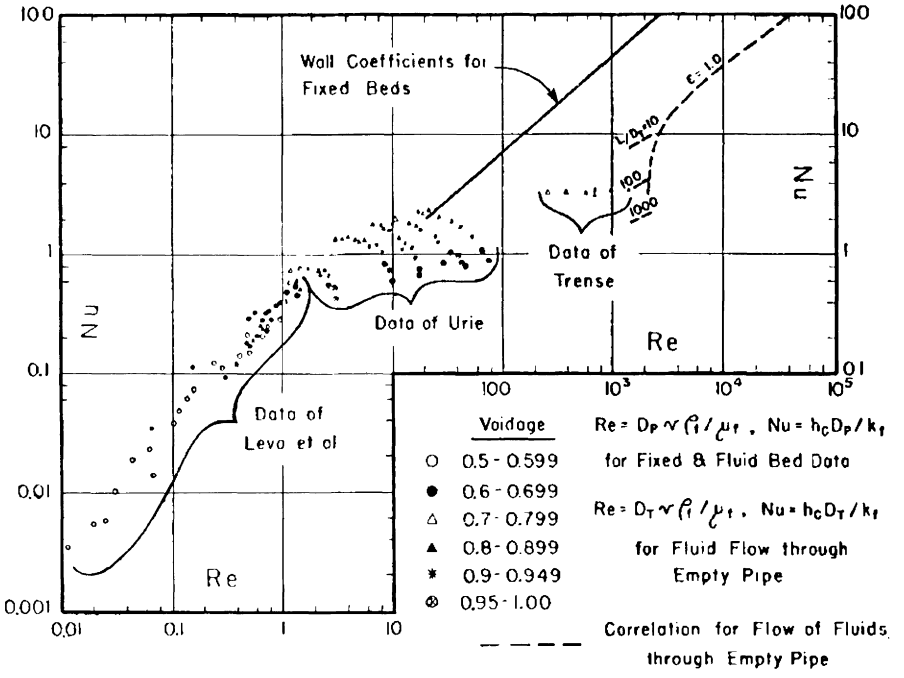
Practically, a particle falling into a fluid bed reaches the bed temperature instantly via conduction heat transfer with other particles. Transfer of heat to the bed from gas is mostly affected by bubble induced bypassing.

Particle to Wall and Tube Heat Transfer 6.3

In a system free of solids, the convective heat transfer term is relatively low due to the stagnant gas layer at the wall. In the presence of fluidized particles, this layer is disrupted. Solids intermittently touch the wall, absorb heat and are replaced by fresh material. The effect is to enormously improve the overall coefficient.

Starting from an unfluidized bed, a sharp increase in heat transfer coefficient is seen as gas velocity passes through minimum fluidization. The coefficient then increases further as the particles become more intensely agitated. Eventually, the density of the fluid bed decreases and the heat transfer coefficient drops as the surface sees more voidage and fewer particles. The heat transfer coefficient also improves as particle size decreases and more surface area and particles are available for heat transport.

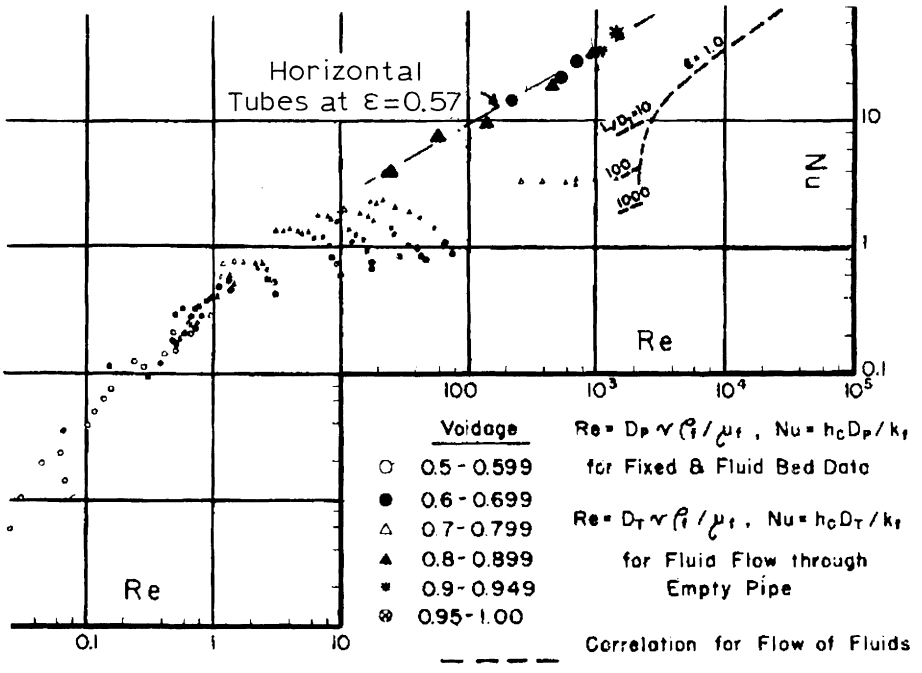
Figure 6.1 is taken from Zenz (2) and presents heat transfer as a function of particle Reynolds number and voidage over a wide range. The figure can be used to give a conservative heat transfer coefficient for vertical tubes and walls in commercial reactors. Similarly, Figure 6.2 (2) applies to horizontal heat transfer systems.



Zenz Correlation for Vertical Heat Transfer Surfaces Fig. 6–1
reprinted from (2)

There are also a number of heat transfer correlations in the literature. They generally provide a reasonable fit to experimental data but no single one stands out as being preferred (3). The correlation of Andeen and Glicksman (4) which is a modification of the Vreedenberg correlation(5) via inclusion of a $(1-\epsilon)$ term can be used. It is,

$$h_c = \frac{900 k_f (1-\epsilon)}{d_t}\left[\left(\frac{G d_t \rho_s}{\rho_f \mu}\right)\left(\frac{\mu^2}{d_p^3 \rho_s^2 g}\right)\right]^{0.326} Pr^{0.3} \qquad [6.3-1]$$



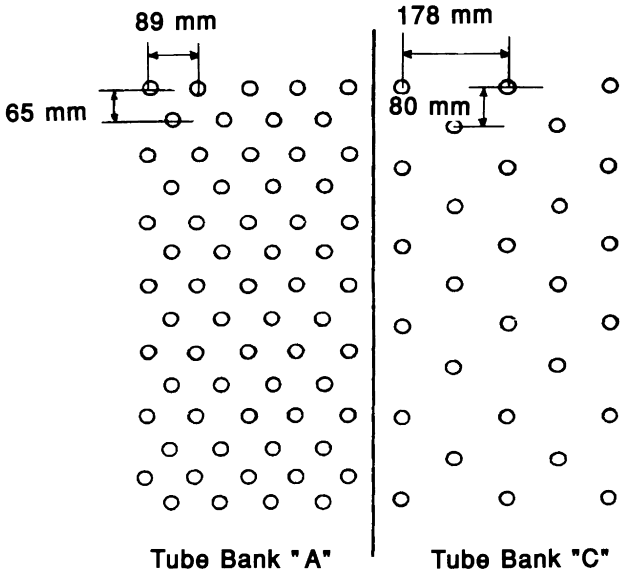
Zenz Correlation for Horizontal Heat Transfer Surface Fig. 6–2
reprinted from (2)

where

G = mass flow rate, kg/m^2 s

Design Considerations for Tube Banks 6.4

In general, it is better to have vertical than horizontal heat transfer surface. Horizontal surfaces are much more subject to poor performance due to hindered circulation (6). The literature provides only limited insight into how much surface can be placed into the bed before problems occur. Further, poor performance of the surface is generally a matter of degree. The system may function but not as well as it was designed to.



Configuration of Tube Banks A and C (7) Fig. 6–3

If the vertical surface occupies no more than 10%–20% of the bed plan area, a conservative design is obtained. At 30%, problems may develop. The solids circulation rate can be considered to be reduced by the fractional plan area occupied by the tubes.

A series of horizontal tubes occupy a greater plan area than a set of vertical tubes. Consequently, performance tends to be poorer since solids circulation is reduced by the baffle effect of the tubes.

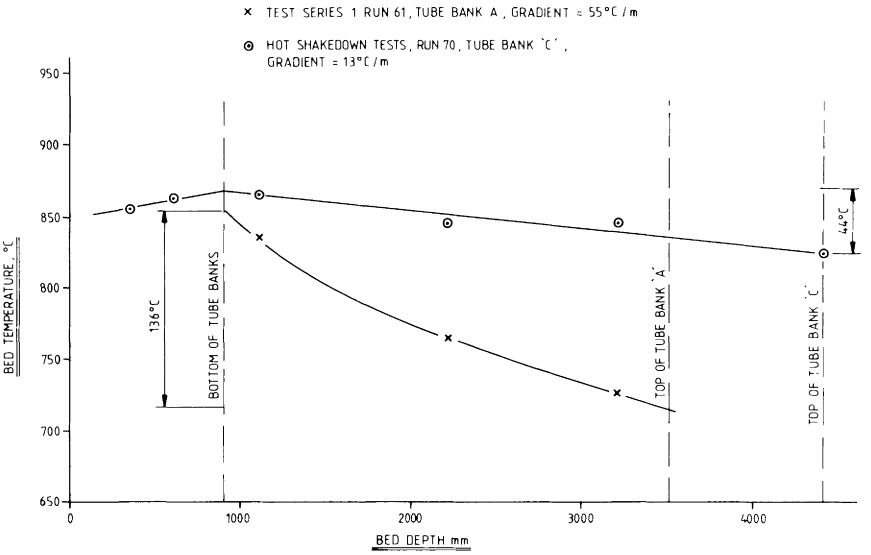
Figure 6.3 and Table 6.1 show two heat transfer banks which were tested in the pressurized fluid bed boiler plant at Grimethorpe, England (7). Temperature drop data are given in Figure 6.4.

The temperature differential across the densely packed tube bank A was 136 °C versus only 41 °C for longer tube bank C. It is possible for the heat transfer coefficient to be according to design and for the overall heat removal to be poor if the particles are not at the assumed temperature. Consider a particle circulating in tube bank A. It is at the nominal bed temperature at the bottom and is moving up slowly due to reduced circulation. Since heat transfer is good, the particle cools as it continues to rise. Its temperature falls and, even though the heat transfer coefficient is high, less heat is transferred due to the reduced temperature

Physical Comparison Between Tube Banks A and C *Table 6–1*

	Tube Bank A	Tube Bank B
Height above distributor, m	0.96	0.99
Height of tube bank, m	2.54	3.44
Horizontal pitch, mm, center to center in the same plane	89	178
Vertical pitch, mm, vertical distance to adjacent row	65	80
Tube OD, mm	32	34
Projected tube area, % of cross section	67	35
Voidage fraction of tube bank	0.86	0.94

Test Series 2.1 Report (7)



Temperature Distribution for Tube Banks A and C (7) Fig. 6–4

difference. The result is that if we make calculations based on an average bed temperature, we see a reduced heat transfer coefficient.

Usually, correlations are obtained under laboratory conditions of good mixing with few tubes present to seriously constrict circulation. In an industrial setting, a lower effective coefficient is a definite possibility.

It is therefore suggested to the designer that horizontal tubes be spaced as open or more open than those of tube bank C. Although this spacing saw some temperature differential, it was relatively slight. For vertical tubes, a limit of somewhere around 20% of the plan area is suggested with 10% being quite safe.

Radiation 6.5

When the bed temperature is less than about 500 °C, the radiation component is less than 15% of the total heat transfer and is usually neglected. Botterill (8) recommends the following for an estimate of the radiation contribution to heat transfer.

$$h_r = 5.673\,E{-}8\;e_r\left(\frac{T_{bed}^4 - T_{surf}^4}{T_{bed} - T_{surf}}\right) \qquad [6.5-1]$$

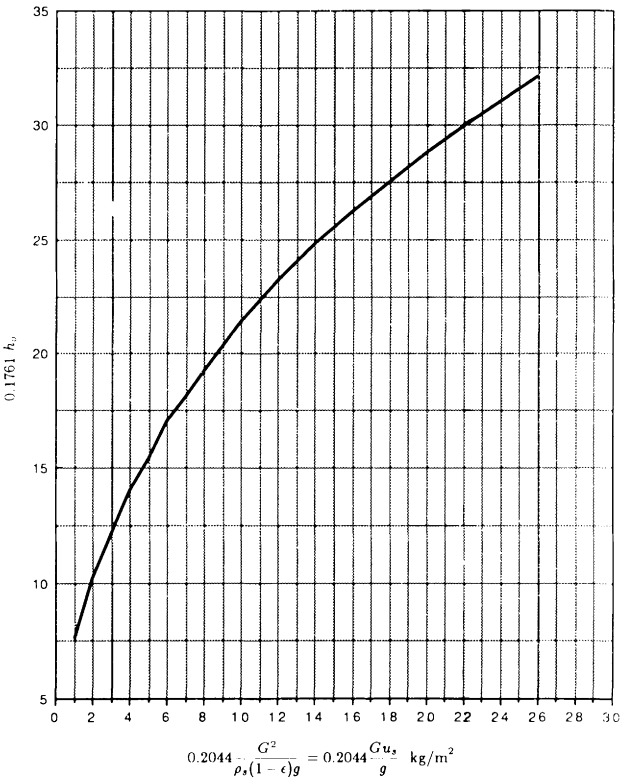
where

h_r is in units of watts/m^2 °K

e_r is the reduced emissivity given by:

$$e_r = \frac{1}{(1/e_{surface} + 1/e_{bed}) - 1} \qquad [6.5-2]$$

The radiation and convection coefficients are added to give the overall fluid bed heat transfer coefficient. An interesting phenomenon associated with the effective bed emissivity exists. Since the particles are opaque to radiation, they can form an effective radiation shield. If we are removing heat from a bed, the particles on the cold surface are close to the cold surface temperature. The surface therefore does not receive radiation from the hot bed (9). For practical purposes, the radiation effects from in-bed radiation heat transfer are negligible at steam raising temperatures. If we are considering heat losses to a hot wall, the radiation component can be quite considerable.



Freeboard Heat Transfer Correlation of Johnson (10) Fig. 6–5

Heat Transfer by Entrained Solids 6.6

There may be heat transfer surface in fast fluidized beds and in the freeboard of fluid beds. For heat transfer in this regime, the correlation of Johnson (10) may be used. The correlation is dependent on the solids throughput or velocity and voidage which is what one would expect from heat transfer by flowing solids.

Particle size is not explicitly given but is present in the solids throughput and voidage terms. Sivakumar et al (11) review particle size effects in upflowing suspensions and show that heat transfer is expected to decline as particle size increases up to about 400 microns followed by an increase in heat transfer with size. The minimum is due to the loss of particle surface area being offset by increase in slip velocity and particle holdup.

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Chapter 7

DRYING AND SOLVENT STRIPPING

Introduction 7.1

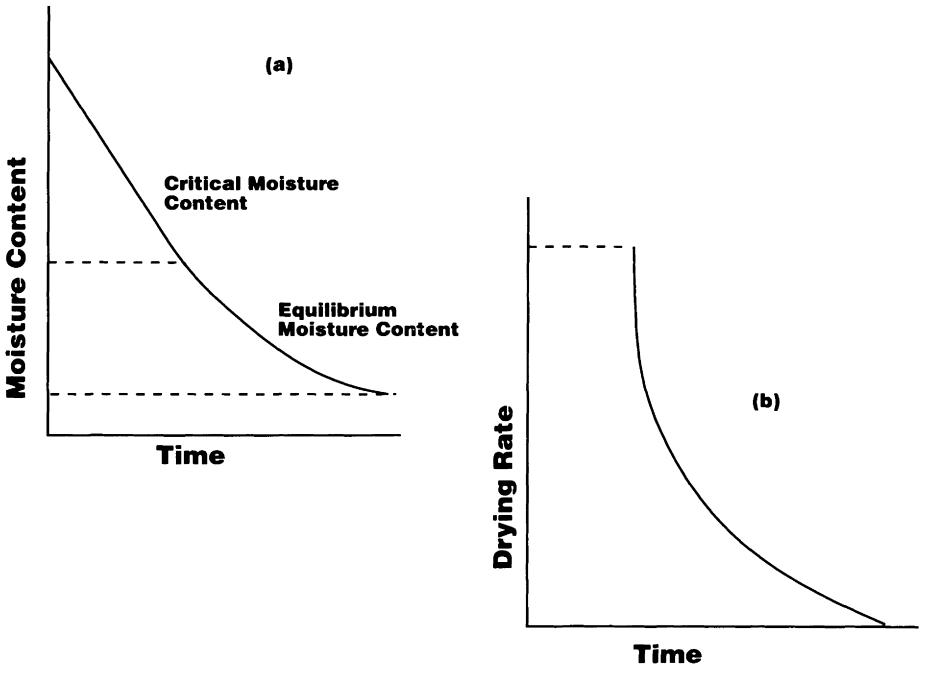
The discussion of drying includes removal of solvents as well as moisture from solid materials. Only continuous operations are discussed. For convenience the term moisture will be extended to include liquid or adsorbed solvent on the solid.

Fluid bed drying of granular fluidizable material gives several advantages over alternative processes. Temperature is uniform throughout the bed so that even product dryness is obtained. There is little chance that sensitive materials will suffer local overheating. The excellent heat transfer coefficient from heating surfaces produces low–cost, minimum surface requirements. Alternately, for heat sensitive materials, the system is suitable for heating with modest temperature differentials.

Fluid beds are backmixed with respect to solids. It is therefore feasible to feed wet or sticky materials into the bed and have them mix in with the dry material so that handling problems associated with wet feed can be avoided.

The Drying Curve 7.2

When solids are dried in a fluid bed, the moisture content typically follows a curve such as that shown in Figure 7.1. Part a shows the moisture content vs time and part b shows the same data expressed as drying rate. A steady decline in moisture vs. time reflects the removal of surface moisture or moisture whose removal is only limited by heat transfer to the particle. This is called the constant rate period. At some point, the critical moisture content is reached. This is the point at which the rate of moisture removal declines. Presumably there are diffusion or desorption effects coming into play which are much slower rate processes than evaporation from the particle surface. Drying curves can vary from being entirely constant rate, to being in mixed

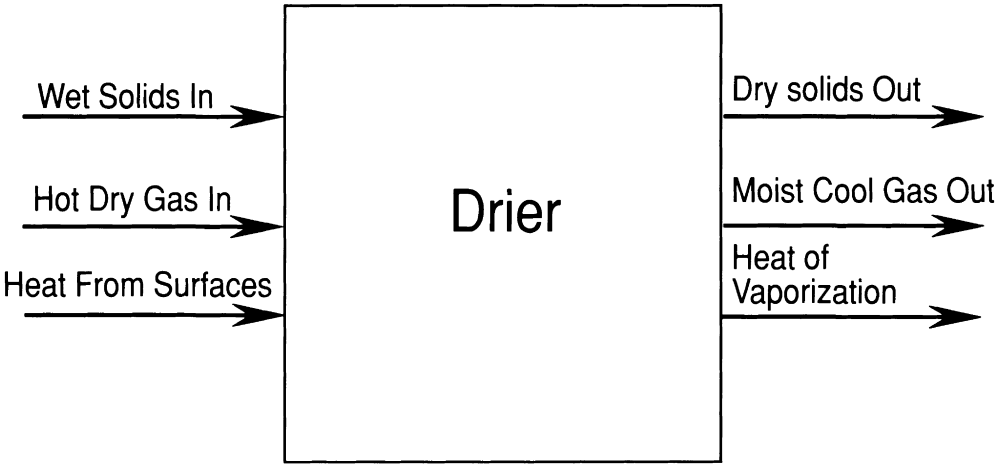


A Typical Drying Curve Fig. 7–1

conditions to being entirely in the falling rate regime. Eventually the particle dries to zero moisture content or to the level in equilibrium with the incoming gas. The curve shows the minimum residence time needed to achieve a desired moisture content.

Design Considerations 7.3

Liquid may vaporize from the surface of a solid by several mechanisms. The first is the constant rate regime. The liquid is present such that there are no diffusion or heat transfer limitations in reaching equilibrium with the gas. Drying is limited by the saturation capacity of the gas and the heat transferred to the solid to vaporize liquid. The situation is shown in Figure 7.2.



Drier Flows Fig. 7–2

The solids quickly reach the wet bulb temperature of the system and the gas and solid exit at the same temperature. A heat balance gives:

$$\textit{heat loss} + \textit{heat of vaporization} = \textit{heat input from surfaces} + \textit{sensible heat given up by solids, moisture and gas}$$

$$Q_{loss} + w_s h_v (X_i - X_o) = Q_{input} + w_f cp_f (T_{fi} - T_o) + (w_s cp_s + w_s X_i cp_x)(T_{si} - T_{so}) \qquad [7.3-1]$$

where

cp = heat capacity
w = dry mass feed rate
X = kg solvent/kg dry solid
s, f, x = subscripts for solids, fluidizing gas, and solvent respectively
i, o = subscripts for inlet and outlet conditions

At the equilibrated temperature of the system, the gas should preferably be some 10 °C above the dew point temperature to prevent condensation in the exit piping. In solvent based systems, an equilibrium curve of the solvent vapor above the solid at the equilibrated temperature is needed to evaluate this condition.

For the regime in which there are no diffusion limitations, design is usually only a matter of getting the needed heat into the overall system. In a diffusion limited situation, the rate process inside the particle requires a certain amount of time and there is less flexibility in design.

The fluidized bed acts as a well mixed vessel and exit solids display a range of particle residence times distributed around the mean residence time. Therefore, individual particles will be wetter and drier than the mean. For some applications, this poses a problem in product quality.

The exit solids residence time distribution is given by (1):

$$E(t) = (1/\bar{t}\,)\,\exp(t/\bar{t}\,) \qquad [7.3-2]$$

where

$E(t)$ = fraction of particles leaving at time t in response to a step change in input.
$\bar{t}$ = average solids residence time.

From the drying curve, the moisture content of the particles as a function of their actual residence time is

$$C(i) = C(t)\,E(t)dt \qquad [7.3-3]$$

where

$C(t)$ represents the moisture content for a given residence time in the bed.

$C(i)$ = fraction of particles having moisture content $C(t)$

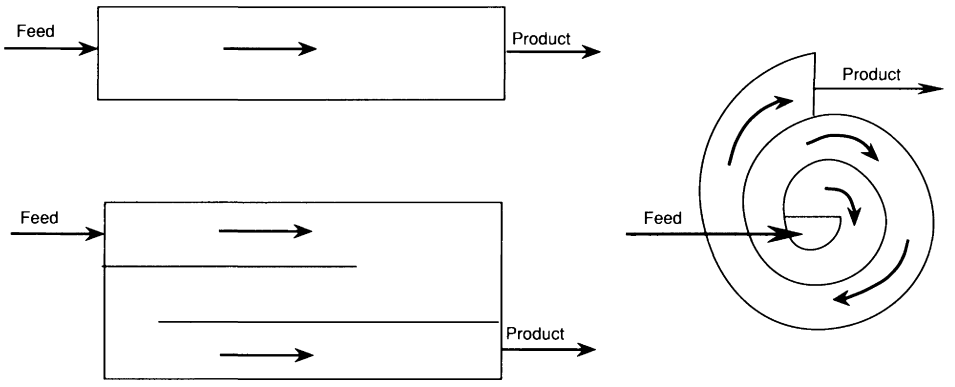
If it is desired to have a specification such as no more than $y\%$ of the particles shall have more than $x\%$ moisture remaining, the single stage residence time can be calculated using the above formulation.

Designing for Residence Time Effects 7.4

In practice, the bypassing characteristic of a well mixed bed makes it difficult to obtain very few wet particles without an inordinate bed volume. The normal practice is to design a "plug flow" drier. This is a unit in which the solids are carried through the system in a flow pattern which minimizes backmixing. Figure 7.3 shows several flow patterns which accomplish this. Figure 7.3a shows a long shallow drier. Since solids must traverse a long path, it is difficult for entering solids to appear at the exit without first spending a reasonable time in the system. Reay (2) gives a method for predicting the staging effect for shallow systems 10 cm or less in height.

The remaining parts of Figure 7.3 show systems in which baffles are used ensure that several stages exist. Modification of the residence time distribution into more than three stages substantially achieves 95% of plug flow and is normally acceptable.

All the systems in Figure 7.3 are crossflow in gas and therefore offer low efficiency in gas usage compared to a countercurrent arrangement. This is especially true for diffusion limited cases in which the exit gas is substantially dilute at the end of the drier. For polymer drying systems, the gas is treated to remove solvent before being recycled or released. In order to make more efficient use of gas, the gas can be placed in a countercurrent arrangement by the commercial system shown in Figure 7.4. As shown, air is first used for cooling the bottom bed. It is then used to combust fuel in hot middle bed. Finally, the warm air leaving the middle bed is used to preheat or dry the feed in the uppermost bed.

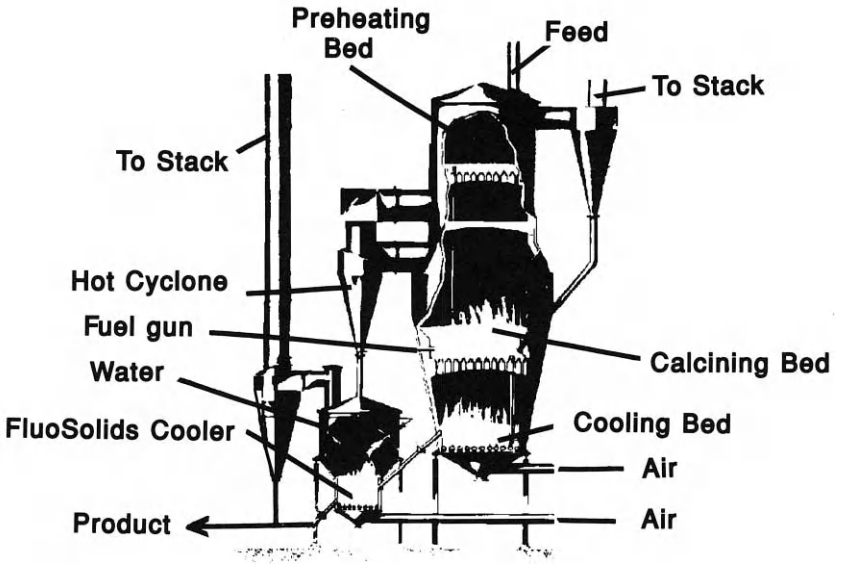


Driers Configured for Plug Flow Fig. 7–3
after (2)

Vibrated Bed Drying 7.5

The minimum fluidization velocity of a system can be substantially reduced by vibrating the bed. The process of mechanically having the bottom of the bed throw the particles up and then having the plate move down more quickly than the particles fall causes gas to be cyclically entrained and expelled from the bed.

Using a vibrated fluid bed drier offers two major advantages. Sometimes the heat and material balance shows that less gas is needed for the drying process than is needed for good fluidization. Vibration can be used to achieve the necessary fluidizing conditions and both gas usage and power consumption can remain low. The other advantage is that there may be large particles or agglomerates in the feed which would otherwise accumulate in the bed. A vibrated system can be set up to use a horizontal component of vibration to move these pieces along the grid plate to the exit end of the drier.



Dorr–Oliver Countercurrent Fluid Bed Arrangement Fig. 7–4
reproduced from (3) with permission

Vendor Assistance 7.6

In many respects, the easiest resolution of a drier design problem is to have a commercial vendor test the material in his lab and/or pilot plant and then obtain a preliminary design and cost estimate. Drying is a well developed business and is so common that performance is often guaranteed by vendors on the basis of their own standardized design procedures.

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Chapter 8

ENTRAINMENT

Introduction 8.1

Entrainment is the carryover of particles from the fluidized bed. In the simplest sense, when the gas velocity exceeds the terminal velocity of a particle, one would expect the particle to be removed from the bed. This is indeed the case. However, when the bed is continually fed with fresh entrainable material, the rate at which particles leave the bed needs to be evaluated. The feed of entrainable material might be from a cyclone so that the entrainable particles are constantly returned to the bed. The feed might be fresh material, or material generated in the bed due to processes of attrition, growth from smaller particles, etc.

Bubbles rise through the bed at a velocity greater than the superficial gas velocity thereby thrusting into the freeboard solids which are coarser than would be expected if terminal velocity vs superficial velocity were the sole criterion for entrainment.

Entrainment calculations are vital to designing the cyclones which must remove the reentrained material. Correlations for reacting systems need the solids concentration in the freeboard. Yet, the literature correlations differ by an order of magnitude on the predicted entrainment and they may suggest opposite directional effects of changes in solid size and density.

Particle Ejection from the Bed 8.2

Pemberton (1,2,3) reviewed mechanisms of particle ejection from fluidized beds. At velocities up to $10 - 15\ u_{mf}$, particles are ejected from the roof of the bubble at a rate of

$$E = 3 d_p \rho_s (1 - \epsilon_{mf})(u - u_{mf})/d_b \qquad [8.2-1]$$

where

E = rate of entrainment, kg/m^2 s

At higher flow rates and for group A particles, bubble coalescence occurs on the surface. The particle ejection rate is given approximately by the size of the bubble wake. By the two phase theory, all gas flow that exceeds u_{mb} goes into the bubbles. Taking the wake as 0.25 of the bubble and assuming that 0.4 of that is ejected gives,

$$E = 0.1\rho_s(1 - \epsilon_{mb})(u - u_{mb}) \qquad [8.2 - 2]$$

The latter equation gives ejection rates over an order of magnitude greater than for the bubble roof mechanism and is likely to be more applicable to commercial processes which tend to stress high throughput applications.

Transport Disengaging Height, TDH 8.3

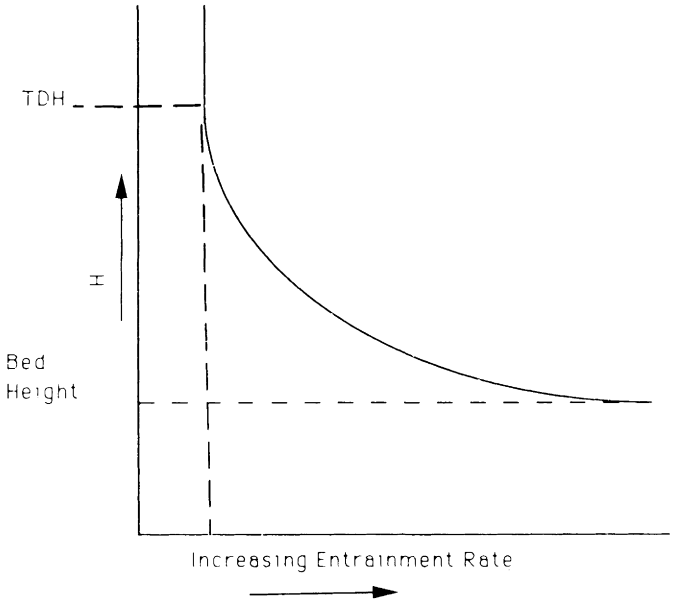
After particles are cast into the freeboard, the coarse ones and clusters of fines fall back as the local high velocities which sent them on their journey are dissipated. Fine particles whose concentration in the gas exceeds the gas carrying capacity tend to collect on the wall and also fall back.

The point at which the entrainment becomes relatively constant is termed the transport disengaging height, TDH. It is usually desired to locate cyclones and vessel outlets well above TDH so as to minimize solids loading in the removal step.

A schematic of the TDH is given in Figure 8.1. It is generally agreed that the solids loading declines in an exponential fashion between the bed surface and TDH.

As might be expected from the mechanism discussed above, TDH is correlated vs. bubble size and gas velocity. The recommended correlation is by Zenz (4) and is given in Figure 8.2.

It is noted that the correlation is oriented toward large systems operating at reasonably high velocities and therefore is likely to reflect the bubble coalescence mechanism rather than the bubble roof particle ejection model.

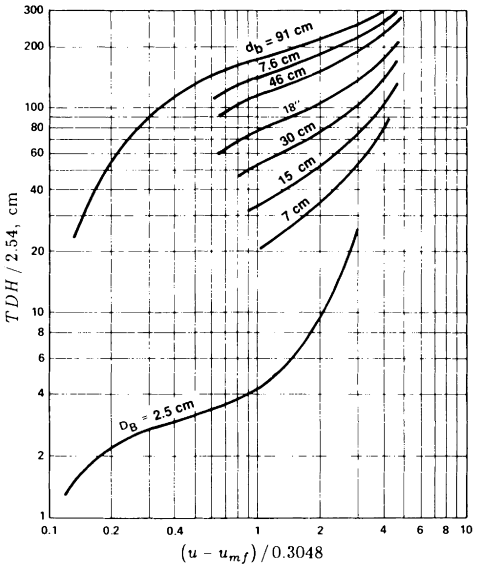


Representation of TDH Fig. 8–1

The above chart does not show an explicit pressure effect. Chan and Knowlton (5) published data on TDH which showed the above correlation to fit well up to 350 kPa (50 psig). But the correlation underestimated TDH as the pressure was raised to 3100 kPa (450 psig).

Correlations for Entrainment Above TDH 8.4

Two correlations are presented below. It is recommended that calculation of entrainment be estimated using both methods and then averaged. The calculation of entrainment is not so well developed as to totally accept any one method over all others at this time.



Zenz TDH Correlation Fig. 8–2

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Geldart's Correlation *8.4.1*

It is universally agreed that the higher the concentration of a given sized entrainable particle in the bed, the greater will be its rate of entrainment. This takes the form of the equation,

$$E(i) = K^*(i)x(i) \qquad [8.4-1]$$

where

$E(i)$ = entrainment rate for size i, kg/m^2 s
$K^*(i)$ = an entrainment constant for particle size i, a function of operating conditions and particle properties.
$x(i)$ = the weight fraction of particle size i.

Geldart (6) recommends using

$$\frac{K^*(i)}{\rho_f u} = 23.7 \exp\left(\frac{-5.4 u_t(i)}{u}\right) \qquad [8.4-2]$$

The composition of the entrainment and the total entrainment rate are then calculated by summing over the individual size fractions.

Zenz's Correlation *8.4.2*

The method given here is an improved version of the Zenz and Weil correlation (7) which is often quoted in the literature. The improved version is given in some detail in a report to the U.S. Department of Energy (8). The fines in the fluid bed are analogous to volatile molecules in a boiling solution. The concentration of particles in the gas is proportional to the concentration in the bed times a relative volatility factor. The concentration of each particle size fraction in the freeboard is determined first. They are then summed to derive the composition of the entrained material. The mean particle size of the entrained material is then used as a parameter for determining the total solids carrying capacity of the gas for the case in question. This then determines the entrainment rate.

The procedure is illustrated via the example given in reference (8). Table 8.1 gives the particle size distribution of the bed being considered. The distribution of the entire bed is not given because only those particles whose terminal velocity is less than the superficial velocity are capable of being entrained. The terminal velocity of each size fraction is calculated and then subtracted from the superficial velocity. Only those particles for which this number is positive are capable of being entrained. This forms columns 1–4.

The dimensionless factor, $(u - u_t)/(SG_p\sqrt{gd_p})$, column 5, is taken as the left side ordinate for the curve of Figure 8.3 and the dimensionless abscissa, $E/(u\rho_f)$, is read from Figure 8.3 and tabulated in column 6. SG_p is specific gravity of the particle defined by the particle density not the skeletal density. The lowest ordinate on the curve has a value of 5. For ordinates less than 5, use the equation:

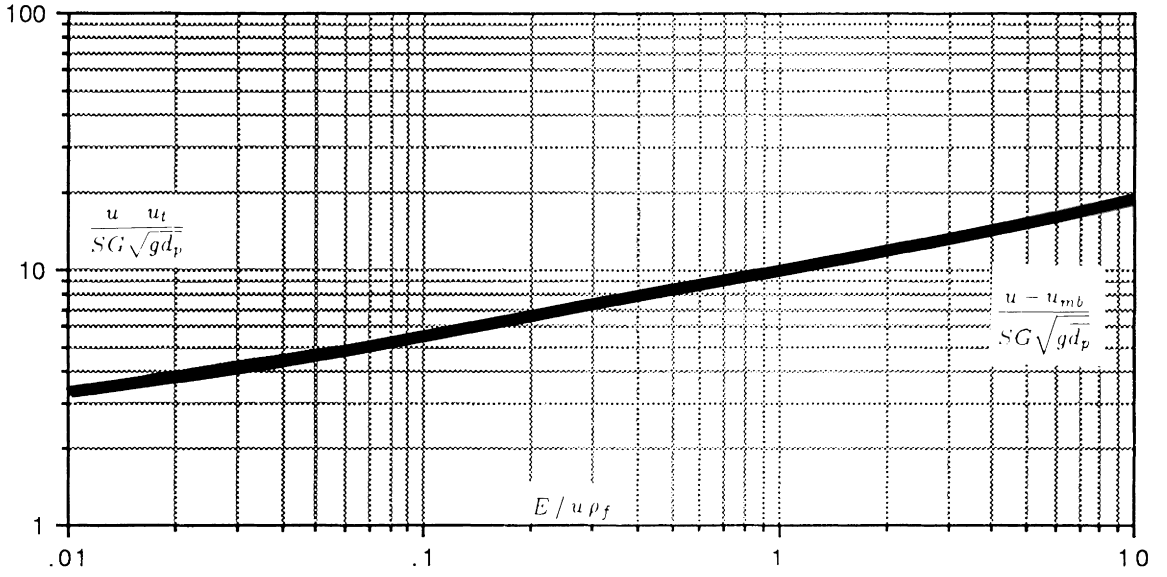
$$abs = 0.0000614\ ord^{4.25} \qquad [8.4-3]$$

where abs and ord = abscissa and ordinate respectively

Sample Estimate of Entrainment via Zenz's Method *Table 8–1*

Bed Comp'n		Velocities		Figure 8.3 Entries		Entrainment Composition		
				Ordinate	Abscissa			
$x(i)$	Avg d_p microns	u_t	$u - u_t$	$\frac{u - u_t}{SG_p\sqrt{gd_p}}$	$\frac{E}{u\rho_f}$	$\frac{x(i)E}{u\rho_f}$	Weight Fraction	Σ Weight Fract'n $< d_p$
0.02	21	0.1104	1.8896	16.69	2.25	0.0450	0.28518	0.0
0.03	29	0.205	1.795	13.49	1.14	0.0342	0.21673	0.2852
0.05	35	0.292	1.708	11.68	0.71	0.0355	0.22497	0.5019
0.10	45	0.465	1.535	9.26	0.30	0.0300	0.19012	0.7269
0.10	55	0.667	1.333	7.27	0.11	0.0110	0.06971	0.9170
0.10	69	0.990	1.010	4.92	0.0196	0.00196	0.01242	0.9867
0.10	85	1.402	0.598	2.63	0.00137	0.000137	0.00087	0.9992
0.10	102	1.875	0.125	0.50	nil	nil	nil	1.000
						0.157797	1.00000	

From F. Zenz (8)



Zenz' Entrainment Correlation Fig. 8–3
from (8)

The dimensionless entrainment factor of column 6 is multiplied by the weight fraction to give column 7. Column 8, the weight fraction composition of the entrainment, is then formed using

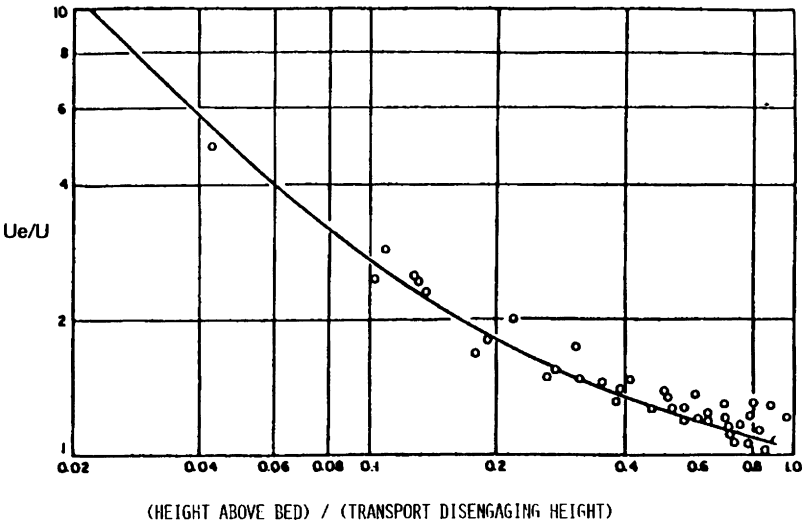
the fractions in column 7. Column 9 is the cumulative weight fraction.

The 50% by weight size based on the column 9 distribution gives d_p for the right side ordinate of Figure 8.3. The right side ordinate is $(u - u_{mb})/(SG\sqrt{gd_p})$. Note that the entrainment is based on superficial velocity less the minimum bubbling velocity. This shows the implicit assumption that entrainment is zero if no bubbles are formed. The abscissa represents the maximum entrainment, that is the entrainment if the entire bed were entrainable. Since only 60% of this bed is entrainable, the abscissa is multiplied by 0.6 to give the final entrainment rate. The composition of the entrainment will be as given by column 8.

Entrainment of Extra Fine and Extra Coarse Particles *8.4.3*

Both methods given above are based upon experiments in which the entrained material does not display significant cohesive behavior. For particles less than 10–30 microns, depending on conditions, cohesive behavior can be important and this will distort the correlations. These fine particles can cling to large particles or to each other. Either behavior will give less entrainment than the correlations predict. Investigation of this phenomenon is at the point where correlations that will account for cohesive particle behavior are not yet available.

It has also been found that very coarse solids with terminal velocities well in excess of the superficial velocity are sometimes entrained above TDH (5,9). The Zenz correlation does not recognize this effect. Geldart et al (9) found that the amount of coarse material entrained was a function of the amount of fines entrained. For a bed containing fine and coarse particles that have very different entrainment characteristics, it is suggested that an effective gas density be calculated as $\rho_f + \rho_p(1 - \epsilon_{freeboard})$ and that this be used to estimate entrainment of the coarse material. The mechanism underlying the entrainment of coarse solids is not yet defined and the calculation suggested above can only be considered a crude guide as to what might be expected.



Entrainment Below TDH Fig. 8–4

Reproduced by permission of the American Institute of Chemical Engineers from (7)

Entrainment Below TDH 8.5

Below TDH, entrainment increases as the bed surface is approached. The Zenz and Weil (7) correlation for entrainment at any distance above the bed and below TDH uses an effective superficial velocity, u_e, as given in Figure 8.4. Bubbles bursting on the surface of the bed eject gas at velocity that is higher than superficial and this gas jetting decays as TDH is approached. Additionally the coarse solids and clusters that fall back to the bed entrain gas down with them and this produces a gas circulation that increases the local upward velocity. Fines moving down the wall also enhance the velocity effect below TDH (2). The correlation uses a velocity that is higher than the superficial velocity to account for these effects.

Care must be taken in evaluating superficial velocity in the freeboard. The presence of cyclone diplegs, heat transfer surface and cyclone bodies all reduce the cross sectional area. If reduced cross section increases velocity before TDH, the entrainment could well be that predicted by the higher velocity rather than that predicted from a lower superficial velocity above the bed.

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Chapter 9

CHEMICAL REACTION MODELING

Introduction 9.1

Commercial scale reactions are typically run in large fluidized beds. Gas velocities are maintained as high is possible to maximize throughput and minimize cost. Under these conditions, many of the models proposed to fit laboratory scale experimental data are inadequate.

The literature contains many fluid bed reaction models but it is beyond the scope of this book to perform a comprehensive review. Instead a model was chosen that is considered to be the most applicable to industrial scale fluid bed reactors.

The model chosen is that of Werther (1, 2). This model was selected for several reasons.

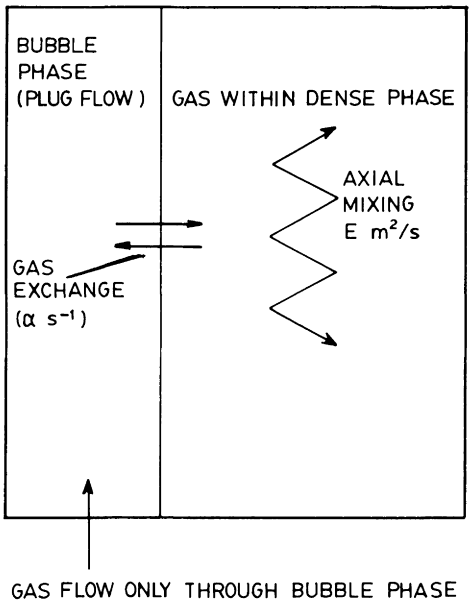
1. The adjustable parameters are based upon observations in fluid beds up to several meters in diameter.

2. It recognizes the existence and importance of a maximum stable bubble size. This enables the impact of the fines content to be incorporated into the model.

3. The model takes into account the effects of bed diameter and of bed height.

4. Geldart group A and group B particles are recognized as being fundamentally different in behavior.

After the bed model is described, the effects of the grid and freeboard will be added.



Two Phase Reaction Model Fig. 9–1

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Historical Background and Evolution of the Model 9.2

In 1959, May (3) studied gas solids contacting in cold fluidized beds up to 1.5 m in diameter by about 9 meters tall at velocities up to 0.6 m/s. The data were expressed in terms of a two phase model as shown in Figure 9.1. Gas entering the bed divides into a bubble phase with no backmixing and an emulsion phase in which there is gas backmixing. A crossflow term accounts for mass transfer between the phases.

The model was refined by van Deemter (4) in 1961. For the design and scaleup of the Shell Chlorine Process, van Swaaij and coworkers (5,6) used the model and refined the parameters related to diffusion control in terms of bed depth, diameter, and fines content. The model was simplified by allowing virtually all the gas flow to go through the bubble phase. It was further shown that, for most industrial cases, the gas backmixing term in the emulsion phase can be neglected.

Werther (1) examined the model and found that an additional term was needed to account for enhanced transfer to the emulsion in the case where chemical reaction was extremely fast. He further extended the model to discriminate between Geldart type A and type B particles.

Description of the Model 9.3

In the bubble phase, concentration change only occurs by way of mass transfer to the emulsion phase. In the emulsion phase, there is diffusion from the bubble phase, reaction, and gas backmixing.

The differential equation for the bubble and emulsion phases under conditions of moderate reaction can be written:

$$\frac{dc_b}{dh} + Na(c_b - c_e) = 0 \qquad \text{bubble phase} \qquad [9.3-1]$$

$$Na(c_e - c_b) - \frac{1}{Ne}\frac{d^2c_e}{dh^2} + Nr\,c_e = 0 \qquad \text{dense phase} \qquad [9.3-2]$$

where

c_b = concentration in the bubble phase
c_e = concentration in the dense or emulsion phase
$Ne = uH/fE_s$, number of backmix stages
E_s = the solids mixing coefficient in the expanded bed, m^2/s (5,6). Here the solids backmix coefficient is used to approximate the gas backmix coefficient.
f = the fraction of the bed occupied by gas in the dense phase
Nr = the reaction resistance defined by kH/u
k = the first order reaction rate constant based on the expanded bed
Na = number of diffusion stages, H/H_α
H_α = height of a diffusional stage, m

Since flow in the bubble phase is usually overwelming compared to flow in the emulsion phase, the bubble phase concentration is equivalent to the overall gas concentration. Although chapter 5 gives data on the solids backmix coefficient and Ne can be determined experimentally if need be, for most industrial units, the Ne term can be neglected. If the backmixing term is small, the equations above resolve to:

$$\frac{C}{C_0} = \exp \frac{-NaNr}{Na + Nr} \qquad [9.3-3]$$

where

C, C_0 = the exit and inlet concentrations respectively.

Conversion is determined by both reaction and diffusion terms. If Nr is large and the rate of reaction is high, then conversion is limited by diffusion. If Na is large compared to Nr, then the reaction is controlled by the rate constant and mass transfer between the bubble and emulsion phases is not a concern.

The key unknown is H_α, the height of a diffusion stage, and how it varies with bed hydrodynamics. Werther's model for H_α (2) encompasses a complex set of factors as given below.

$$\begin{aligned} H_\alpha &= 0.05\phi_b(d_t)F1(H,h^*)F2(u-u_{mf}) && \text{group B powders} \\ &= 0.015\phi_a(d_t)F1(H,h^*)F2(u-u_{mf}) && \text{group A powders} \end{aligned} \qquad [9.3-4]$$

The effects of bed diameter and particle type are given by:

$$\begin{aligned} \phi_{a\ or\ b} &= \phi' && d_t <= 0.1 \text{ m} \\ &= \phi'' d_t^{0.4} && 0.1 \text{ m} <= d_t <= 1.0 \text{ m} \\ &= \phi'' && d_t => 1.0 \text{ m} \end{aligned} \qquad [9.3-5]$$

where

$\phi' = 1.0$ for group A and 0.64 for group B particles

$\phi'' = 2.5$ for group A and 1.60 for group B particles.

The effects of bed height and bubble growth are contained in the function, F1. F1 is given separately for bed height less than and greater than that corresponding to the maximum stable bubble size.

For $H > h^*$,

$$F1(H,h^*) = \frac{H}{0.18(1-(1+6.84H)^{-0.815}) + (1+6.84h^*)^{-1.815}(H-h^*)}$$

For $H <= h^*$,

$$F1(H,h^*) = \frac{H}{0.18(1-(1+6.84H)^{-0.815})} \qquad [9.3-6]$$

Where h^* = the height at which bubble growth ceases or the height at which the maximum stable bubble size is attained. As discussed in Chapter 4, this is a function of the fines content, velocity and distributor. Finally,

$$F2(u-u_{mf}) = [1+27.2(u-u_{mf})]^{0.5} \qquad [9.3-7]$$

The above equations are suitable when conversion is not extremely high and are suitable to most industrial applications. However, the model assumes gas transfer through a film surrounding the bubble. When the reaction is very rapid, reaction in the film is sufficient to reduce the reactant concentration and thereby enhance the mass transfer. The Hatta number squared is the ratio of material convertible in the film to material transported through it and therefore explicitly defines the enhancement phenomenon. Using analogy to gas liquid systems, Werther (1) obtained for first order reaction an expression which reduces to equation 9.3–3 above at Hatta numbers << 1.

$$C/C_0 = \exp -\frac{(1/\Phi-1)Ha + \tanh Ha}{(1/\Phi-1)Ha\ \tanh Ha + 1}\ HaNa \qquad [9.3-8]$$

where

Φ = the ratio of film volume to volume of the emulsion phase.

The following are needed to define the dimensionless quantities Φ and Ha.

$$Ha = \frac{\sqrt{kD_{eff}}}{\alpha'} \qquad [9.3-9]$$

where

k = the reaction rate constant based on emulsion phase volume, or $(1 - \epsilon_b)$ as defined below.

D_{eff} = the effective gas diffusivity. The gas diffusivity may be substituted for the effective diffusivity (1).

α' = the mass transfer coefficient per unit area, 0.0088 m/s for group B solids and 0.0159 m/s for group A particles.

$$\Phi = \frac{aD_{eff}}{(1 - \epsilon_b)\alpha'}$$

Where $a =$ interfacial area per unit volume, $a = \dfrac{u - u_{mf}}{\alpha' H_\alpha}$

We define ϵ_b as the fraction of bubble volume and this is given by:

$$\epsilon_b = \frac{C1(u - u_{mf})F3(H, h^*)}{\phi_{A\ or\ B}(d_t)[1 + 27.2(u - u_{mf})]^{0.167}} \qquad [9.3 - 10]$$

where

$C1 = 2.47$ for group B and 3.27 for group A particles.

We then have the function related to bubble growth and maximum bubble size, $F3$.

$$F3 = 0.37((1 + 6.84H)^{0.4} - 1)/H \qquad H =< h^*$$

$$= 0.37((1 + 6.84H)^{0.4} - 1)/H + (1 + 6.84h^*)^{-0.6}/H \qquad H > h^* \qquad [9.3 - 11]$$

The above relationships are fairly complex but they describe a complex set of hydrodynamic relationships that are the best model for commercial systems that the author is aware of.

Effect of Fines 9.4

For maximum reactor efficiency, we would like H_α to be as small as possible. The height of a mass transfer stage, H_α, increases with fluid bed height and then remains steady at the height at which the maximum stable bubble size is achieved. In Chapter 4, both correlations for maximum stable bubble size are a function of the fines, −44 micron, in the bed. DeVries et al (5) gave data relating fines directly to their observed H_α. H_α decreased with fines up to 30%. Pell and Jordan (8) presented data from a pilot sized unit which showed conversion increasing markedly with increasing fines up to 44%, however, the effects on selectivity were not discussed. Squires et al (9) published a figure which implies an optimum fines level on the order of 25%. At some level of increasing fines, the system will move away from group A particles into the poor fluidization characteristics of group C particles and reactor efficiency will deteriorate.

Unfortunately, the effect of fines on maximum stable bubble size will not be seen until the operating system is large enough to encounter that phenomenon. The bed height and diameter needed may not be available until pilot plant size is attained. Nonetheless, the effect is extremely important in full sized reactors; variation in fines may therefore have to be deferred as an experimentally investigated variable until the commercial unit is available.

It has been reported that increasing temperature decreases H_α (6) and increasing the pressure has little effect. Changing velocity has be said to have little effect on H_α. However, Pell et al (8) found that increasing velocity decreased H_α and Werther's model has H_α increasing with increasing velocity. Clearly, the velocity effect remains to be defined better.

Grid Region Effects 9.5

The model given above does not separate different parts of the reactor for separate treatment. A key area to be considered is the grid region because there is extensive evidence of increased reaction due enhanced mass transfer where gas enters the reactor. Consider the case of gas forming a bubble under a mass of particles. The only way in which the solids can be moved out of the way to form a bubble is for a force to be exerted by gas flowing past the solids. Indeed it has been found that the gas

observed in the initially formed bubble is between 0.3 and 0.5 of the expected $Q - Q_{mf}$ (10,11). Most of the gas then finds its way back into the bubble to complete its passage through the bed. The resulting flow into and out of the emulsion phase enhances the mass transfer. In the case where jetting occurs, there is also intensely enhanced mixing due to convection effects versus normal diffusion effects.

The enhancement in mass transfer at the grid versus the bed is an order of magnitude or greater (12). In this case, the mass transfer resistance of the grid region becomes very small and the reaction can become rate limited. A simple way of taking this into account is to assume a divisor on H_α estimated from Behie and Kehoe (12) of about 10. The height of the grid region can be taken as the jet length penetration given in Chapter 3. Na for the grid region is then:

$$Na_{grid} = 10\frac{l_j}{H_{\alpha_{bed}}} \qquad [9.5-1]$$

where

l_j = jet length as found in Chapter 3.

The height of the grid region is used to calculate Nr_{grid}. Then, $C/C_{0\ grid}$ is calculated using equation 9. The height of the bed is reduced by the height of the grid region when calculating the bed contribution to C/C_0.

This gives an method for separately taking into account the grid region effect. For fast reactions, it correctly predicts that a significant extent of reaction takes place at the very bottom of the bed. In addition, it helps explain why the addition of bed height may yield very little improvement in overall conversion.

A problem with the above method is that it does not give quantitative prediction of the effects that have been observed when the grid configuration is changed (13). In addition, the H_α correlation given above was derived for the entire bed including the grid region and it is not clear how the correlation should be affected by considering the grid region separately. Still, it is one way to account for changes in conversion that may be associated with changes in grid configuration.

Bauer et al (14) used a detailed interfacial area calculation to model the effects of different grid configurations and found that, for the system tested, grid effects could be explained without having to change the value of H_α.

Further study of grid region conversion using commercial grid configurations and especially avoiding sintered metal plate grids is needed.

Freeboard Region 9.6

It is well known that a significant portion of the catalyst inventory of a reactor may be present in the freeboard rather than in the fluid bed (15,8). In the freeboard, bubbles are not present and gas forms the continuous medium rather than the solids.

Yates and Rowe (16) developed a model for freeboard reaction which estimates the material ejected from the bed and then compares the mass transfer rate with the reaction rate assuming good dispersion of the solids. Following the nomenclature of Wen (17) the freeboard conversion can be expressed as:

$$C/C_{0\ freeboard} = \exp -(Nr_{freeboard}) \qquad [9.6-1]$$

where

$$Nr_{freeboard} = k' h_{freeboard}/u$$

and

$$k' = \left[\frac{d_p(u-u_t)\rho_s}{6k_g E} + \frac{1}{ke'}\right]^{-1} \qquad [9.6-2]$$

Here we have used the entrainment rate and entrainment size estimated from Zenz's correlation as in Chapter 8. e' represents the ratio of the solids concentration in the freeboard to the concentration referred to the kinetic constant, usually ϵ_{mf} or the fixed bed voidage. Therefore e' is $(1-\epsilon_{freeboard})/(1-\epsilon_{mf})$ and the freeboard voidage is given as $1 - E/[(u-u_t)\rho_s]$.

where

E = the entrainment rate expressed as kg/m^{2}s

k_g can be estimated from:

$$\frac{k_g dp}{D} = 2 + 0.69 \frac{\mu}{\rho D} Re_t \qquad [9.6-3]$$

where

D = the gas diffusivity

Using the above procedure, one can calculate freeboard conversion by using the single value of entrainment above TDH. A more rigorous alternative is to go stepwise along the freeboard including the zone below TDH and substituting the appropriate values for entrainment and particle size.

The total conversion can be given by:

$$\frac{C}{C_0} = \prod_{i=grid}^{i=freeboard} \frac{C}{C_0}(i) \qquad [9.6-4]$$

where

i = grid, bed, freeboard sections as needed.

Baffles 9.7

Baffles may or may not be effective in assisting conversion in the bed. For group B particles in systems for which bubble growth is significant, it is possible to keep the bubbles small by using baffles (18). However, for group A solids, the bubbles are usually small enough to not be radically affected by the baffles unless the baffles are spaced sufficiently close to interfere with overall solids circulation. It is then possible to get voids and recirculation zones around the baffles. Although baffles are recommended as helping reactor performance (19), their effectiveness will depend on the details of baffle design and the resulting hydrodynamic changes in the reactor.

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Chapter 10

SCALE UP AND SCALE DOWN

Introduction 10.1

We build large scale commercial units for production and we build small scale units for gathering data which will enable us to understand system parameters. We use the small scale results to design and operate the large commercial unit for maximum efficiency. The most valuable information will be obtained if, to the greatest extent possible, the small scale unit is designed from the beginning as a scale down of a commercial unit rather than as the most convenient lab or pilot scale unit.

Early work on a chemical reaction processes is usually done on a very small scale in packed beds or some form of micro reactor. This is suitable for catalyst screening and for preliminary kinetic studies. For the industrial researcher, a laboratory scale fluid bed is preferred to a packed bed. Use of a packed bed to acquire data for a fluid bed reactor scaleup should be avoided since gas and solids backmixing effects have excellent potential for changing the observed kinetics. This is especially so when adsorption effects are present since the particles alternately are transported between high conversion and low conversion zones of the bed. The combination of a changeable solids environment with gas backmixing is likely to affect yields. The fluid bed, even in small scale, gives the first indication of potential bubble mass transfer limitations.

The Scaleup Sequence 10.2

Lab Scale *10.2.1*

The laboratory unit is the first step in the scaleup process. We are not discussing catalyst or feedstock screening as these can usually be done on a micro scale or in fixed beds. We are interested in the response to various process parameters and what the predicted yields are. How big should the lab unit be? As large as possible within the constraints of the facility. Five to fifteen centimeters in diameter is a good range.

The grid should simulate a commercial system if at all possible. The use of a sintered metal plate gives excellent gas distribution and the smallest initial bubble size and therefore tends to produce optimistic data. A grid based on a commercial design such as a drilled plate or bubble cap is best. The configuration can then be varied to give insights into the sensitivity of the process of changes in the grid region.

The combination of bed height, gas velocity and diameter should be adjusted to avoid slugging. This can be difficult for small beds. Based on the initial bubble size, the procedures of Chapter 4 can be used to check for slugging. The bed should be designed for the greatest height and velocity consistent with process investigative needs.

The bed performance should be evaluated using the analyses shown in Chapter 9 for kinetic and diffusional resistances. Usually diffusional resistance is minimal in the lab unit. But the diffusional resistance must be factored into design calculations for conversion as the scaleup sequence progresses.

Cold Model *10.2.2*

Usually a cold model of the system in the next size scale is constructed. Sometimes a full scale model of the commercial unit is also constructed. The cold model can be used to provide tracer data on diffusional resistance, backmixing and adsorption effects (1, 2, 3). The cold model is inevitably cheaper than the hot unit that it simulates and may give insights into system properties such as:

1. Jet penetration
2. Effects due to placement of internals
3. Location of inlets and outlets
4. Solids mixing in horizontal and vertical directions
5. Minimum fluidization and bubbling velocity, bed voidage
6. Gas distribution
7. Entrainment
8. Attrition
9. Gas diffusion and bypassing
10. Heat transfer coefficients

It is best to build a three dimensional cold model but much useful information can be obtained from a thin slice approach where the bed is only several centimeters thick. Care must be taken to evaluate what the wall effect might be on the variable of interest in two dimensional thin units.

How closely does the cold model simulate hydrodynamics of the hot unit? What are the effects of low pressure vs. high pressure operation? Glicksman (4,5) developed a scaling relationship from first principles for going from cold to hot and from low pressure to high pressure for group B particles. The relationship holds the following ratios should be held constant to maintain hydrodynamic similarity: gd_p/u^2, h/d_p, d_t/d_p, ϕ_s, particle size distribution, bed geometry and a final term which is $\rho_s^2 d_p^3 g/\mu^2$ for viscous dominated flow and ρ_f/ρ_s for inertial dominated flow.

For an intermediate condition, both last terms should be kept constant. Viscous flow dominates when the particle Reynolds number, $\rho_f u \phi_s d_p/\mu$, is 4 or lower and inertial flow dominates at $Re > 400$. This leads to the conclusion that, for systems like fluidized bed combustion, the hot unit might be in inertial flow while a cold model might be in viscous flow. However, running the cold model to achieve the Glicksman scaling criteria led to operation of the cold bed with iron grit of 25% the particle diameter of the silica hot bed. It may be difficult to go to such lengths for the simulation. Also, it is not yet universally accepted that this is necessary.

For group A and group C materials, interparticle forces have been neglected in the above fundamental analysis. We do not yet have a similar scaling guideline for these particles. Most people simply operate the bed at the same velocity as the hot bed. Others try to adjust particle size so the minimum fluidization velocity is the same in the cold unit as in the hot unit.

Pilot Plants *10.2.3*

The ideal pilot plant operates, without slugging, at the same velocity of the commercial plant and is large enough so that maximum stable bubble size is reached. For catalytic systems, this is generally achieved with a diameter of 45–60 cm. In pilot scale units, the effects of fines on maximum stable diameter can be seen and the solids should be sized to match the commercial unit.

The recommended procedure for scaleup is to model conditions according to the methods given in Chapter 9. This takes into account effects of bed diameter and height and is based on commercial experience.

For group B solids, there may be no choice but to allow slugging in the pilot plant. When the pilot plant operates under slugging conditions, Zenz (6) gives a simplified method to determine the height of a scaled up bed as:

$$h/h_{pilot} = 1.68(d_b/D_{pilot}) \qquad [10.2-1]$$

where

d_b = estimated average bubble diameter in the scaled up unit.

Another possible scaleup rule is offered by Horio et al (7). They recommend that the small scale unit which has a diameter $1/m$ of the larger unit be set so that:

$$(u - u_{mf})_{large} = \sqrt{m}\,(u - u_{mf})_{small}$$

and

$$u_{mf\ large} = \sqrt{m}\,u_{mf\ small} \qquad [10.2-2]$$

Furthermore the large to small ratios of bed height, hole diameter and hole pitch should all be *m*. This will produce geometrical similarity of bubble properties between the two units.

Large Scale Effects 10.3

Occasionally, effects occur for which we do not yet have easy predictors. Vibration and shaking may begin in a large vessel when there was no indication of the problem at all in the small scale. Chapter 13 addresses some vibration issues related to bubble size. There may be large scale distribution effects. For example, feed points may need to be made multiple where only a single one was needed in the pilot plant. Some discussion of radial diffusivity was made in chapter 5 but our predictive techniques still need improvement. As size increases, gas distribution in the plenum may have to be modified. Waves and currents could conceivably be set up in units where there is a large recirculation from cyclones. Thermal expansion problems can suddenly appear that were insignificant in the small scale. These types of problems are not easily predicted. Very often they are only discovered after the large unit is built. When the cause of the difficulty is not obvious, it can sometimes be found via large scale cold model testing.

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Chapter 11

DEFLUIDIZATION AND AGGLOMERATION

Introduction 11.1

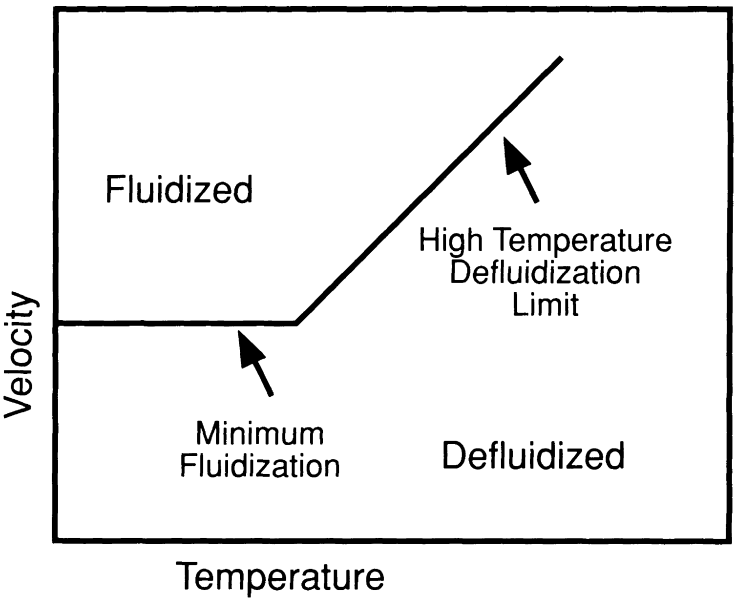
Solids in fluidized beds can become sticky in several ways. An added liquid phase such as water or residual solvent may be present. A component of the solid may become liquid during processing; this can occur with coal or ore solids. Polymers may become sticky as the operating temperature nears the melting point. Sometimes we are operating a fluid bed with the deliberate addition of a liquid phase to induce agglomeration or as a reagent.

If the solids stick together, the particle size increases and the bed could defluidize. This chapter will be largely qualitative because widely applicable design procedures for predicting agglomeration and preventing defluidization are not well developed.

Defluidization from Thermal Sintering 11.2

Siegell (1) gives an excellent review of thermal sintering as it influences defluidization. The materials that Siegell investigated included copper shot, glass beads, polymers and coal. At u_{mf}, the force of gravity is balanced by drag from upflowing gas. When the surface is sticky, an additional force is needed to separate the particles before fluidization can occur. This force can be provided by additional drag from higher velocity.

The most important variable affecting thermal stickiness is temperature. Below the initial sintering temperature, T_s, the bed will not agglomerate even though it is not in motion or fluidized. Once the sintering temperature is reached, the bed will agglomerate at u_{mf}. Above T_s, the excess velocity needed to maintain fluidization, $u - u_{mf}$, is proportional to the temperature, $T - T_s$. As the melting point is approached, the required velocity rises exponentially and eventually the bed defluidizes as the solid melts



Velocity vs Temperature Effects Fig. 11–1
redrawn from (1)

no matter what the velocity. The phenomenon is shown graphically in Figure 11.1

It is important to recognize that the sintering temperature can be very different from the melting point. This means that a fluid bed can suffer thermal defluidization at a surprisingly low temperature. Compo et al (2) list a number of sintering and melting temperatures.

Rather than running a fluid bed test, it is possible to run a lab scale dilatometry experiment in which the expansion of a small sample is monitored as temperature is increased. As the sintering point is approached, the expansion starts to level off. Eventually contraction occurs as the melting point is approached. The maximum in the expansion vs. temperature curve corresponds to the initial sintering temperature (1). Compo et al (2) examined a number of crystalline materials. They found that, while the behavior above usually holds, some crystals continue to expand even above T_s.

Siegell (1) reported that as bed depth increased, the sintering temperature was reduced. This was probably due to the increased load on the particles. Compo et al (2) showed that increasing the load on the sample lowered T_s.

It was also shown that as particle size decreased, the sintering temperature also decreased. This is to be expected since there is an increase in contact points available for interparticle forces to be applied. The attractive force, which is from point contact for two spheres, is independent of size. But it is a greater fraction of the force balance for small particles than for large ones.

Pressure drop through the fluidized bed also changes as the bed begins to defluidize. Usually, as the bed solidifies, the gas flow maintains a channel into which the agglomerated bed particles cannot fall. This clear channel gives an extremely low pressure drop. The channeling induced low pressure drop is an excellent instrumental indication that a bed is defluidized. At low multiples of u_{mf}, the bed may solidify without any channels being formed. In that case, the pressure may be very much higher than for the unagglomerated bed.

Normally, a higher velocity will keep the bed fluidized over a greater range of conditions. However, if a high enough level of turbulence is applied (especially for fine particles) it is possible to compress and deform the particles thereby encouraging agglomeration. Also the energy being input can locally raise the temperature of the particle surface which also encourages agglomeration. Thus there are industrial pieces of equipment which deliberately achieve agglomeration through the application of intense turbulence.

It was noted above that finer particles are more subject to interparticle forces than larger ones. If the particles are fine enough, agglomeration will occur without the presence of a liquid phase. This effect is usually seen only when the particle diameter is less than about 10 *microns*.

Intentional Agglomeration 11.3

Many industrial products are produced in agglomerated form in fluidized beds. Very fine particles are difficult to handle in terms of metering, packing, application, loss, environmental control etc. Intentional fluid bed agglomeration is practiced for pharmaceuticals, herbicides, pesticides and other products. Typically the material is processed by adding a liquid to the bed in the form of a fine spray.

Maroglou and Nienow (3) reviewed the various process mechanisms. There may be binder in the liquid spray or the solids in the bed may have binder characteristics. The agglomeration process begins as the spray droplets first form liquid and then solid bridges between particles. Depending on the relative effects of liquid bridge strength, dry bridge strength, fluidization velocity, relative liquid addition rates, etc. the product characteristics are determined. The system may also be set up to produce a product which is formed by accretion, or drying of the liquid spray solution on the bed particles.

The literature is not so well developed as to assure design of a product without experimentation. There is a considerable body of proprietary information involved and certainly much experimental work before a process design is assured. A number of equipment vendors offer pilot and lab scale facilities for rental (4) and will usually assist with design of the agglomeration step.

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Chapter 12

ATTRITION IN FLUIDIZED BEDS

Attrition Fundamentals 12.1

Attrition is the reduction in size of particles in the fluidized bed. We can measure the size of the bed particles over a period of time and calculate the rate of loss per unit weight of bed particles. Since the size reduction generates a corresponding weight of finer material, the attrition rate can also be given by the fines generation. This is particularly convenient if the fines leave the bed and are collected externally.

When a particle experiences an impact, cracks can propagate which result in cleavage of the particle and loss of material. More simply stated, when the particle is hit hard enough, something breaks. Furthermore, breakage occurs first at the weakest point. In distinguishing between particles, the weakest ones are the most likely to break. Two characteristics of attrition follow from the above. The first is that particles of irregular shape containing numerous asperities or sharp edges will attrite at those asperities first since the smaller cross sections provide points of stress concentration. One therefore expects a high initial rate of attrition from such particles followed by a reduced rate as the material becomes rounded.

The other expected phenomenon is that the weakest particles will break or attrite first. Either natural or manufactured particles will have a certain strength distribution. The weak ones will break first and those that are left will be the stronger particles. This phenomenon also leads to a declining rate of attrition with time.

A key question is whether the attrition process is one of particle breakage into similar sized pieces or whether the particle is ground down to give fine material coming from the exterior with the particle size slowly decreasing. The latter phenomenon is what we generally deal with in fluidized beds. This is an abrading or attriting process rather than a crushing process. The fines

produced from an abrading process tend to have the same size distribution independent of the initial particle size (1). The material is considered to have a "natural grain size" which is produced when abraded.

There are numerous mechanisms which can cause a particle to decrepitate in a fluid bed operation. Vaux and Keairns (2) review a number of these and they include such things as change of crystal structure due to chemical reaction, thermal shock due to sudden heating, conveying and feed devices, etc. While any of these can occur, we will concentrate on the more usual source, impact breakage.

Attrition in fluidized beds is a complex process with a number of mechanisms capable of being in operation. Attrition is approximately proportional to the square of the local velocity to which the particle is subjected. In a commercial fluidized bed, the velocity in the bed is on the order of a meter per second or less. The grid jet velocity is on the order of thirty meters per second and a first stage cyclone inlet velocity is on the order of twenty meters per second. Thus the greatest sources of attrition are likely to be the jets or the cyclones.

Time Effects on Attrition 12.2

Measured attrition rates decrease with the time the particle is in the bed. The attrition rate is usually measured as the amount of fines generated per unit time per mass of particles. With this definition, we would count as attrited material fines entering the system with the feed or in the system when it is started up. This is not really attrition but might be counted as such since we measure the appearance of fines independent of their source. In running an attrition test, this can account for a very fast initial attrition rate. In evaluating attrition in a continuously fed fluid bed, the fines in the feed should be subtracted from the overall attrition rate.

The next fastest fines generation step comes from rounding of the particles. Sharp edges and protrusions are broken off first and particles tend to become rounded as they remain in the bed. This gives an attrition rate that is initially quite high until the particles are rounded.

Finally even the round particles will have a distribution of strengths so that some will be more susceptible to attrition than others. Although the attrition rate may appear constant relative to the fast changing rates due to initial fines loss and the rounding of edges, the attrition rate will continue to change when viewed on a longer time scale. If a catalyst bed starts up with fresh material and experiences an attrition rate of 0.2% per day, it will take 500 days to achieve one bed replacement. It would take four years to reach the traditional three turnovers needed for 95% equilibration of the bed.

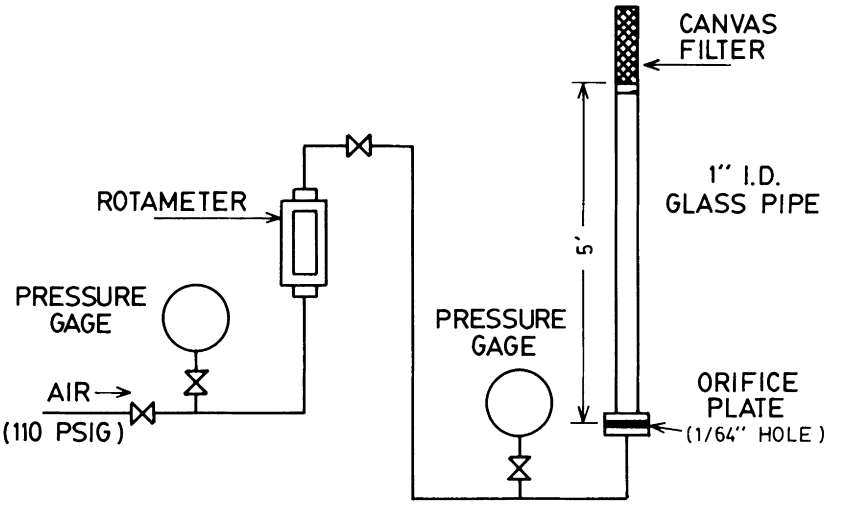
Attrition Tests 12.3

One can illustrate the difficulties of picking an attrition test via a simple thought experiment (3). If we took a batch of rubber pencil erasers and a batch of diamonds, and rubbed them on abrasive paper, we would conclude that the diamonds were more attrition resistant. If we instead struck the particles with a hammer we would conclude that the rubber erasers were more attrition resistant. Ideally an attrition test should duplicate the controlling attrition mechanism in the fluid bed of interest.

A huge variety of attrition tests have been quoted in the literature (4,5) and certainly there are many more in use which are proprietary and have never been published. All of these tests give the relative attrition resistance of two or more materials. Prediction of performance in a process is difficult unless material that gives known attrition rates in the commercial unit is available.

Fluid Bed Tests *12.3.1*

As mentioned above, the fluid bed itself is usually not the greatest source of particle attrition since higher velocities are encountered in the grid jets and cyclones. If a sintered plate or another means of introducing fluidizing gas at a low velocity is used and there is little solids carryover into cyclones, then the limiting attrition rate might indeed occur in the bed. In this case, an attrition test could consist of a fluidized bed operating at the same velocity as the commercial application bed. The test bed should be made as deep as possible so as to most closely duplicate the bubble action and bed weight of the commercial unit.



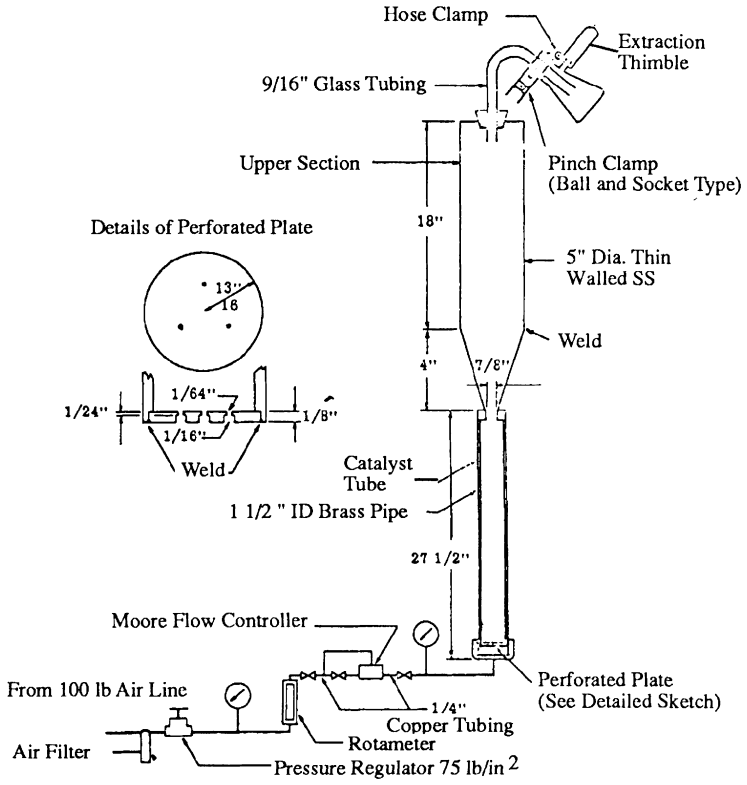
Apparatus of Forsythe and Hertwig (6) Fig. 12–1

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Jet Attrition Tests *12.3.2*

The jets can cause more attrition than the bed motion. There are numerous tests which involve a jet in a fluid bed. Figures 12.1 and 12.2 show two simple jet attrition setups that are often quoted in the literature. Figure 12.1 shows the apparatus of Forsythe and Hertwig (6). A single jet is centered in an orifice plate and the system is run at a velocity of about 800 m/s. The test is run for an hour and the particle size is measured before and after. The apparatus of Gwyn (7) shown in Figure 12.2 is similar in function except that three holes are used and fine material is removed as the test progresses. A jet velocity on the order of 300 m/s is used and the test is run for a number of hours. The expanded section is sized so as to allow cracking catalyst particles larger than about 20 microns to be elutriated. Attrition is measured as the percentage of these particles collected.

Neither test will directly yield design information, but rather gives relative attrition resistance under conditions designed to simulate those in a commercial unit.



Apparatus of Gwyn (7) Fig. 12–2
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Hardgrove Grindability Index *12.3.3*

There is no one attrition test method that is accepted as being superior to others. However, there is a highly standardized test method that is widely used and whose value is frequently reported in the literature. This is the Hardgrove Grindability Index, ASTM D–409 (8). If the attrition process of interest is dominated by grinding, it is likely that different materials will attrite in the order of their HGI. The test requires grinding a presized sample and then measuring the fines. A low number corresponds to few fines and a harder to grind particle.

Cyclone Tests *12.3.4*

Zenz and Kelleher (9) describe a cyclone attrition test in which the material is passed batchwise through a small cyclone at commercial velocity a number of times. The fines are collected and blended back with the cyclone catch after each run. Data are presented but not a correlation suitable for predictive purposes. Considering that attrition in the cyclones is likely be the dominant mechanism for many processes, it is surprising that the literature doesn't contain more tests based on cyclones or cyclonic action.

Design Procedures 12.4

The preferred design procedure is to be in the position of knowing the attrition rate of standard fluid bed material and then comparing the relative rates of attrition using a fluid bed or jet attrition test as described above. The attrition test should be chosen so as to match the controlling mechanism in the commercial bed.

Bed Attrition *12.4.1*

Vaux and Schruben (10,11) quote an attrition rate, R, due to a bubbling bed as:

$$\frac{RH}{u - u_{mf}} = \frac{[F(t) + 1]g\rho_s H^2}{g_c \sigma} \qquad [12.4-1]$$

where

R = attrition rate, kg/kg s
$F(t)$ = an exponentially decreasing function of time or extent of attrition
σ = the fracture energy of the solid.
H = bed height

Equation 12.1 shows that attrition increases with bed depth. Ray et al (1) question whether the bed attrition is indeed proportional to bed height and this issue still remains to be resolved. For fluidized particles in a bed with constant bubble size, one should not expect a big effect of height. The experiments of Vaux et al were conducted with coarse limestone. Bubble size would

increase with bed height and much more of a bed height effect is expected.

Grid Attrition *12.4.2*

Zenz and Kelleher (9) correlate grid jet attrition with the expression that has attrition proportional to the number and area of the jets times the jet velocity to the 2.5 power:

$$\text{Attrition, gm/min} = C(up_f)^{2.5} N \frac{\pi}{4} d_o^2 (1 - F) \qquad [12.4 - 2]$$

where

C = a numerical constant given below for SI units.
F = the weight fraction of fines in the bed.

With the attrition rate defined as less than a particular d_p produced, C is given for silica-alumina fluid cracking catalyst as:

Attrition Rate Constants for Silica Based Catalyst *Table 12–1*

Size $0 - d_p$, microns	Numerical Constant for Downflow Jets in Equation 12.4-2
0–2	0.0011
0–23	0.00903
0–50	0.0228

Zenz and Kelleher (9)

The above constants are for downward pointed jets. For upward pointed jets or flat plate grids, the constants are divided by about 2. Glass beads are found to give constants about 1/12 of those for silica based cracking catalyst (9).

The equation above provides insight into anticipated effects of varying conditions but does not give an absolute prediction of attrition. One could run an attrition test on the material of interest and on a silica alumina cracking catalyst to get an estimate of predicted jet attrition. But without the original catalyst used to develop the table, even that procedure would give uncertain results.

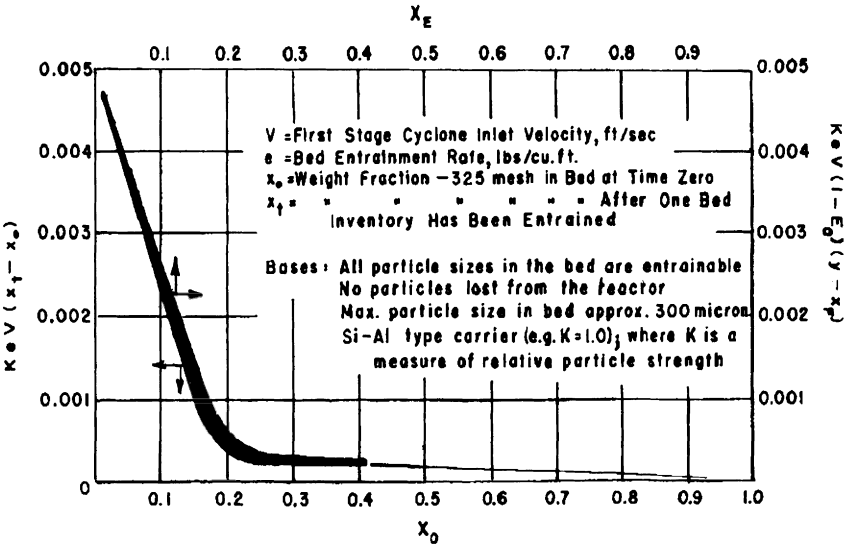
Note that downflow gives a higher attrition rate than upflow. Presumably the downflow jet finds it more difficult to push particles out of the way when those same particles are circulating upwards counter to the local gas jet. Similarly, it is important to check the jet length to ensure that no internals are in the way of the jet or particle attrition and erosion of the internals will both take place. Similarly, care must be taken to prevent two opposing jets from approaching closely.

Cyclone Attrition *12.4.3*

Extensive data for predicting cyclone attrition in commercial practice are generally not available. However, Zenz (12) presents a calculation curve based on measurements from a single reactor. The calculation procedure requires use of the graph in Figure 12.3 which is given in terms of English units. The rate of attrition is derived from the cyclone efficiency which determines which particles are retained in the system. Fine particles cannot contribute to attrition since they are taken as the attrition product. The cyclone inlet velocity affects the impact forces and a term is included for particles which are different from the silica alumina based catalyst from which the data are derived. The catalyst makeup rate is equal to the total loss rate. This loss rate includes fines generated from all attrition processes plus the fines that are fed in. The calculation procedure has many assumptions and is based on limited data. Additionally there is no discussion of the contribution due to grid attrition.

Designing to Minimize Attrition 12.5

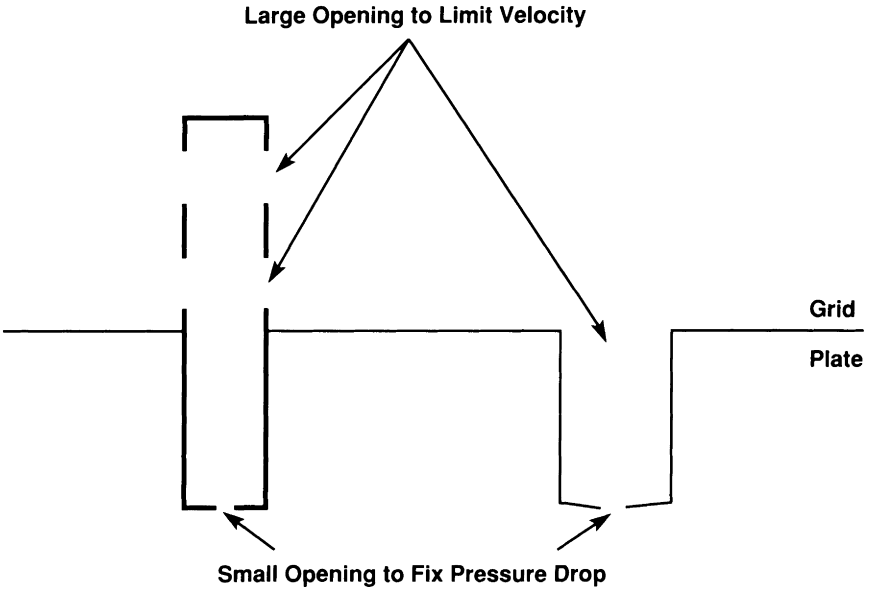
Grid attrition can be minimized by reducing the velocity at which gas enters the bed. As discussed in chapter 3, we want a certain velocity through the grid holes in order to have sufficient pressure drop to ensure good gas distribution. This velocity often is enough to produce a high attrition rate. US Patent 3,974,091 (13), discussing fluid cracking regenerators, teaches that the gas velocity seen by the bed should be reduced by means of a shroud or a pipe of larger diameter concentric to the hole that provides the required pressure drop. The sketch of the system in Figure 12.4 is the same one shown in chapter 3. The covering pipe must be long enough to contain the gas jet until it has expanded enough to reach the wall. The gas jet is assumed to expand at an included angle of 11 degrees and the covering pipe is made



Impact of Cyclone Attrition (12) Fig. 12–3
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long enough to allow the jet to fill the pipe. An exit velocity of less than 40 m/s is recommended but this will depend on the attrition resistance of the bed material. Another method is to use a bubble cap in which the pressure drop is taken upstream of the exit holes.

One method of minimizing cyclone attrition is to limit the quantity of material entering the cyclone. Foremost is the need to have enough height so that the cyclone is above TDH. Another design technique is to use a freeboard with larger diameter than the bed. The gas velocity is reduced and more particles fall back into the bed without reaching the cyclone. The first stage cyclone can be conservatively designed to have a low inlet velocity on the order of 15 m/s. The secondary cyclones are unlikely to contribute significantly to the overall attrition because typically 99% of the material entering the cyclones is removed in the first stage. Additionally, material entering the secondary cyclones is mostly fines and therefore does not contribute to attrition. These techniques are extensively practiced in industry.



Grid Design to Minimize Attrition Fig. 12–4

If the fines loss rate of an existing bed is known, deducting the rate of fines feed yields the production of fines via attrition. Equation 12.2 above gives the loss rate due to the grid and the remainder is therefore due to the cyclones. This evaluation shows where the attrition is coming from and can aid in redesign of the unit to further reduce losses.

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Chapter 13

PHYSICAL FORCES

Effect on Particles 13.1

Large bubbles rising through a fluid bed have several physical force effects associated with them. One already alluded to is the attrition effect on particles in the bed. The physical force on particles leads to attrition and breakage. As suggested in Chapter 12, this is usually not the greatest source of attrition. However, the effect can be major in some systems. Kono et al (1) have investigated the force on large particles 0.5-3 mm in diameter in a system 0.15 m diameter by 1.5 m high. In general, the force increased with bed height and particle size. Depending on velocity, the largest particle experienced a force of up to 0.3 N.

Effect on Internals 13.2

Turner and Irvin (2) investigating forces on tubes in a cold rig, found that forces increased with bed height and velocity up to 2.0 m/s and declined at higher velocity. Several investigators (2,3,4) have found that a power spectrum analysis (amplitude of force vs frequency) shows that forces on internals tend to be a maximum at a frequency of about 1.5 Hz and that there is little amplitude above about 20 Hz.

Hosny and Grace (5) found forces on the order of 30 N/m of tube. They investigated the effect of tube size, particle density and particle size. Their mechanistic model says that a tube being passed by a bubble will experience four periods of changing forces associated with the gas filled bubble and the particles in the wake of the bubble alternately streaming by. Forces were separated into horizontal and vertical. The net horizontal force was zero but the RMS value was about a fifth of the vertical force. The model applies to group B particles where the velocity is not high enough to change the bubble regime.

Each time period is given a fraction ϕ_i.

$$\phi_1 = 1 - \phi_2 - \phi_3 - \phi_4 \qquad [13.2-1]$$

where

ϕ = fraction of the time between passing bubbles
subscripts denote the four different periods

In the first period, the tube is not influenced by bubbles and only experiences a buoyant force. The force consists of the buoyant force due to the tube volume being immersed in the bed with a factor to account for the stagnant material on top of the tube. Thus

$$F_{v1} = 0.7\rho_s(1 - e_{mf})g\,V_{tube} \qquad [13.2-2]$$

where

F_{v1} = vertical force during time period 1.
V_{tube} = the volume displaced by the tube.

There is no horizontal component to the buoyant force and:

$$F_{h1} = 0$$

where

F_{h1} = horizontal force during period 1.

In the second period, bubbles are below the tube. The time fraction is, $\phi_2 = f d_b/u_b$ where f is the bubbling frequency. Hosny and Grace (5) used $2.2/s$, constant, for f. The forces are now proportional to the tube area seen by the bubble and the kinetic energy of the moving solids. Using empirical constants they get forces of

$$F_{v2} = \rho_s(1 - e_{mf})u_b^2 A_{tube}$$

and

$$F_{h2} = 0.2\rho_s(1 - e_{mf})u_b^2 A_{tube} \qquad [13.2-3]$$

The third period is the time that the tube is immersed in the bubble. The time fraction is the fraction of bubbles in the bed and the force is zero in both directions. Then:

$$\phi_3 = \epsilon_b = \approx 0.6(u - u_{mb})/u_b$$

and

$$F_{h3} = F_{v3} = 0 \qquad [13.2-4]$$

In the fourth stage, the tube is influenced by bubbles above the tube surface up to the expanded bed height so that:

$$\phi_4 = f_b(h - h_{tube} - d_{tube}/2 - d_b/2)/u_b \qquad [13.2-5]$$

Empirically, the forces are given by:

$$F_{v4} = F_{v1} - 0.065F_{v2}$$

and

$$F_{h4} = 0 \qquad [13.2-6]$$

The forces are summed to give the RMS forces as:

$$F_{v \text{ or } h}RMS = \sqrt{\sum_{i=1}^{4} \phi_i F^2_{v \text{ or } h\ i}} \qquad [13.2-7]$$

The net horizontal force will be zero and the net vertical force will be in the up direction. At low velocity, the vertical force will be proportional to tube volume; at higher velocity, this will change to tube area facing the flow. The horizontal forces will be less than 20% of the vertical forces. Effects such as gulfstreaming which is seen in large vessels are not included.

It is important that internals be designed with a natural frequency well away from the several Hz found to be the predominant frequency range in bubbling beds.

The above is only a first look at forces on internals in the bed. The effect of momentum change as the bubble wake strikes the tube needs to be added. A large scale circulation pattern will have a horizontal component with its associated momentum effects. Downcomers from cyclones or circulating fluid beds will also produce momentum changes which need to be accounted for.

Effect on the Vessel and Associated Equipment 13.3

The problem of vibration and forces on the vessel containing the bed and the attached equipment can be substantial. The difficulty becomes worse as the bed becomes deeper, the particle size larger and the particle more dense. Consider a bed containing solids several hundred microns in diameter with a density of 3,000 kg/m^3. If the bed is deep, bubbles of substantial size can be formed. A 2 meter bubble has a wake volume on the order of $0.4\pi 2^3/6$ or 1.7 m^3. As the wake is ejected from the bed surface, it returns with a weight of 5,000 kg. I have stood on platforms tied to deep beds of dense material and have felt the bed and the entire structure vibrate as if being struck by a several thousand kg force. This impact will travel through the vessel and cause the vessel and the supporting structure to resonate at their own natural frequencies. The effect is similar to striking a tuning fork; the impact from the surfacing bubble causes the structure to vibrate at its own natural frequency. The advice of this section is therefore to be aware of these potential problems when designing systems which potentially can grow bubbles large enough to throw large amounts of mass about.

References 13.4

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Appendix A

UNITS AND SYMBOLS

SI Base Units and Prefixes A.1

Quantity	Unit	Symbol
length	meter	m
mass	kilogram	kg
time	second	s
temperature	kelvin	K
chemical substance	mole	mol

Prefix	Multiplier	Symbol
mega-	10^6	M
kilo-	10^3	k
centi-	10^{-2}	c
milli-	10^{-3}	m
micro-	10^{-6}	μ
nano-	10^{-9}	n

Derived SI Units with Special Names A.2

Quantity	Unit	Symbol	Equivalent	Base Equivalent
frequency	hertz	Hz	Hz	s^{-1}
force	newton	N	N	kg m s^{-2}
energy	joule	J	N m	kg m^2 s^{-2}
pressure	pascal	Pa	N/m^2	kg m^{-1} s^{-2}
power	watt	W	J/s	kg m^2 s^{-3}

Conversions from Non-SI Units A.3

Unit	Symbol	SI Equivalent
degree	°	0.017453 rad
liter	L	10^{-3} m^3
erg	–	10^{-7} J
dyne	–	10^{-5} N
poise	p	0.1 Pa s

Some units that are not in either the cgs and SI systems remain in common use. These are related to SI units as follows:

Unit	Symbol	SI Equivalent
Angstrom	Å	10^{-10} m
micron	μm	10^{-6} m
gram-calorie	cal	4.184 J
millimeter of mercury	mm Hg	133.3 Pa
pounds-force/inch^2	PSI	6.895 kPa
atmosphere	atm	101.325 kPa

Universal Constants A.4

Constant	Symbol	Value
acceleration of gravity	g	9.8066 m/s^2
Planck constant	h	6.6262 10^{-34} J/s
Boltzmann constant	k	1.3805 10^{-23} J/K
ideal gas law constant = $N_0 k$	R	8.3144 J/(mol K)
pi	π	3.1416

Symbols used infrequently are defined in the text near the equations in which they appear.

Roman Font A.5

Symbol	Meaning	Units
A	area	m^2
a	interfacial area per unit volume	1/m
abs	ascissa	–
Ar	Archimedes No. $d_p^3\rho_f(\rho_p - \rho_f)g/\mu^2$ the same as the Galileo No.	–
C	concentration, kg moisture/kg carrier for drying calculations	–
C	numeric constant for attrition rate Chapter 12	–
c	concentration for reaction calculations	moles/m^3
cp	heat capacity	watts/kg °K
D	diffusion coefficient	m^2/s
d	diameter, for particles sometimes expressed in microns, for vessels and internals and bubbles, always in meters	m
d_s	arithmetic average based on screen size	m
d_v	diameter of a sphere whose volume is equal to that of the particle	m
d_{sv}	surface volume average diameter, $1/\sum x_i/d_i$	m
dp	pressure drop	kPa
E	entrainment rate, Chapter 8	kg/s m^2
E	solids axial diffusion coefficient, Chapter 9	m^2/s
e_r	reduced emissivity, Chapter 6	–
e_{bed}	emissivity of bed, Chapter 6	–
e_{surf}	emissivity of surface, Chapter 6	–
$E(t)$	fraction of particles leaving the system as a function of time	–
F	weight fraction <45 micrometer size in the bed	–
	force in Chapter 13	N
$F(t)$	a decreasing function with time, Chapter 12	–
F1,2,3	functions used for determining Ha and $H\alpha$ in Chapter 9	–
f	weight fraction bubbles in the bed	–
f	weight fraction of bed occupied by gas in the dense phase, Chapter 9	
f	frequency in Chapter 13	1/s

Symbol	Meaning	Units
g	acceleration of gravity, 9.8 m^2/s	m^2/s
G	mass flow per unit area	kg/m^2s
Ga	Galileo number, see Archimedes number above	–
H	fluidized bed height	m
h	bed height, height above grid	m
Ha	Hatta number, square root of the ratio of material convertible in the bubble film to that transported through the film	
H_α	height of a diffusion stage	m
h_c	convective heat transfer coefficient	watts/m^2 °K
h_r	radiation heat transfer coefficient	watts/m^2 °K
k	first order reaction rate constant	1/s
k_g	mass transfer coefficient	m/s
k_f	fluid thermal conductivity	w/m °K
l	length	m
M	mixing index, $x/\bar{x}$	–
m	diameter ratio for scaling	–
N	number of items or number of subscripted resistance stages for reaction and diffusion	–
N	Number of items	–
Na	number of diffusion resistance stages, $\alpha H/u$ or $H/H\alpha$	–
N_e	number of axial diffusion stages, uH/fE	–
N_r	number of reaction stages, kH/u	–
Nu	Nusselt number, $h_c d_p/k_f$	–
n	exponent in the Richardson–Zaki equation	–
ord	ordinate	–
p	pitch, hole center to hole center	m
Pr	Prandtl number, $c_p\mu/k_f$	–
Q	gas flow rate	m^3/s
R	attrition rate	g/g s
Re	Reynolds number based on channel diameter and fluid properties, $d_T u\rho_f/u$	–
Re_p	Reynolds number based on particle diameter and fluid properties, $d_p u\rho_f/u$	–
RMS	root mean square	–
SG	specific gravity ρ_s/ρ_{water}	–

Symbol	Meaning	Units
t	time	s
T	temperature	°K
T_s	initial sintering temperature, Chapter 11	°K
tan	tangent	–
tanh	hyperbolic tangent	–
U	Overall heat transfer coefficient	watts/m^2
u	superficial velocity, as if the bed were empty	m/s
V	volume	m^3
w	flow rate	kg/s
X	weight of solvent per weight dry solid	–
x	weight fraction	–
$\overline{x}$	weight fraction of sinkers in the bed, Chapter 5	–
z	velocity ratio for mixing, Chapter 5	–

Greek Font A.6

Symbol	Meaning	Units
α	gas diffusion coefficient	1/s
α'	mass transfer coefficient per unit area	m^2/s
δ	term for solids properties, $(3\mu^2/4\rho_f(\rho_p - \rho_f)g)^{1/3}$	m –
ϵ	voidage	–
ρ	density	kg/m^3
Φ	bubble film volume/emulsion volume, Chapter 9	–
ϕ(A or B)	function for dependence of H_α on reactor diameter, Chapter 9	–
ϕ	time fraction under the influence of a bubble, Chapter 13	–
π	the constant 3.14	–
μ	viscosity in consistent units or centipoise	cp
σ	fracture energy of a solid	dynes/cm
ω	term for gas properties, $(4\mu(\rho_p - \rho_f)g/3\rho_f^2)^{1/3}$	m/s

Subscripts A.7

Symbol	Meaning
a	axial
b	bubble
cf	complete fluidization
d	drift, describes solids following a rising bubble but not in the wake
e	emulsion phase
eff	effective
fl	floaters or top layer, chapter 5
f	fluid
H	high density component
h	horizontal
i	counting index referring to a given fraction inlet when referring to flow
j	jet
L	low density component
l	liquid phase
mb	minimum bubbling
mf	minimum fluidization
min	minimum
o	orifice or grid hole referring to diameter, out when referring to flow, outside for heat transfer coefficient
p	particle
r	radial for diffusivity, radiation for heat transfer
s	solids
si	sinkers or the bottom layer
S	saltation
sc	screen size
t	terminal velocity
T	tank, vessel, pipe or tube
TO	takeover, refers to velocity at which mixing starts to predominate, chapter 5
v	volume in Chapter 2 vertical in Chapter 13
w	wake of the bubble
x	solvent
0	initial condition

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HANDBOOK OF POWDER TECHNOLOGY

Vol. 1. Particle Size Enlargement, by C.E. Capes

Size enlargement methods (the techniques used to transform fine particles into granular materials) are used in industries ranging from mineral processing to food and fertilizer production. This book presents the first concise generalized account of different methods used in these industries, with emphasis on the underlying principles and unifying characteristics of each method. Information is presented concisely in tables, diagrams and figures whenever possible, together with the minimum amount of descriptive text necessary. In the introductory chapter, guidelines are laid out to help in the choice of a suitable size enlargement method for new applications. Finally, a comprehensive reference list is provided with each chapter to allow further study in greater depth.

Vol. 2. Fundamentals of Gas-Particle Flow, by G. Rudinger

This monograph treats the analysis of flowing suspensions of solid particles in a gas. The text is organized in sections which deal with the behavior of single particles in various flows, the thermodynamics of gas-particle mixtures, steady flow of suspensions with negligible or finite particle volume, considering both equilibrium and non-equilibrium conditions, and the propagation of shock waves, sound waves and waves of arbitrary finite amplitude. The book also discusses topics such as gas-particle jets, boundary layers and applications of gas-particle momentum exchange. Examples, many worked out in detail, are provided to illustrate a variety of flows, and a brief section on experimental techniques is included. Cross references are given, where appropriate, to other volumes of the Handbook where topics are treated in more detail.

Vol. 3. Solid-Gas Separation, by L. Svarovsky

This volume provides a concise, yet highly technical account of existing gas cleaning technology. Drawing on his many years of involvement with the subject, the author has aimed at providing a balance between descriptive material and basic theoretical concepts needed for the design and understanding of the equipment considered in the book.

The book concentrates on the principles and technology of gas cleaning but in addition, three associated topics are dealt with, namely, the relevant dimensionless groups, the efficiency of separation, and the economics of gas cleaning.

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Vol. 6. The Packing of Particles, by D.J. Cumberland and R.J. Crawford

The wide range of information currently available on the packing of particles is brought together in this monograph. The densification of a powder mass is of relevance in a great many industries, among them the pharmaceutical, ceramic, powder metallurgy and civil engineering industries. In addition, the packing of regular - or irregular - shaped particles is also of relevance to a surprisingly large number of other industries and subject areas, i.e. the foundry industry, nuclear engineering, chemical engineering, crystallography, geology, biology, telecommunications, and so on. Accordingly, this book is written for a very wide audience.

The authors have treated the subject in as logical a manner as possible. Hence, after an introduction describing some early work in the field and the types of packing which occur in nature, there is a chapter on the packing of finite areas with circles. After this two-dimensional case, there is a chapter on the three-dimensional situation where a volume is packed with spheres. Initially, mono-size spheres are considered and then the complexity increases as binary, ternary, quaternary, quinary, and n-component systems are introduced. Then follows a chapter on practical considerations in particles packing, and this includes some observations on future developments in this subject area. Considerable efforts have been made to collect as much relevant information as possible on each of the subject areas and, accordingly, an extensive bibliography is provided.

Vol. 7. Dispersing Powders in Liquids, by R.D. Nelson

This book provides powder technologists with laboratory procedures for selecting dispersing agents and preparing stable dispersions that can then be used in particle size characterization instruments. Its broader goal is to introduce industrial chemists and engineers to the phenomena, terminology, physical principles, and chemical considerations involved in preparing and handling dispersions on a commercial scale.

The book introduces novices to: industrial problems due to improper degree of dispersion; the nomenclature used in describing particles; the basic physical phenomena, equations, and chemistry involved in particle interactions; the analytical techniques used for evaluating solid-liquid interfaces; textbooks, courses, societies, and vendors that can provide an advanced understanding of dispersion phenomena.

The book provides resources for more experienced technologists by: discussing characteristics and applications of the various chemical classes of surfactants; providing procedures for selecting and optimizing a dispersant for a specific solid-liquid system and end-use constraints; naming typical commercial surfactants and listing the addresses and telephone numbers of their manufacturers; illustrating the many physical and chemical equilibria that must be considered in modelling a dispersion and guiding the reader to the sources of both data and advanced theoretical treatments required to implement such models.